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LLMs and IoT: A Comprehensive Survey on Large Language Models and the Internet of Things

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ABSTRACT The Internet of Things (IoT) has emerged as a technological cornerstone, enabling highly interconnected systems that integrate billions of diverse devices. The combined data output of these devices is estimated at hundreds of exabytes per day. The rise of Large Language Models (LLMs) presents new opportunities for managing these devices, processing the data they generate, and addressing the technological and societal challenges posed by IoT ecosystems. However, the integration of LLMs into IoT also introduces significant challenges. This article offers the most comprehensive survey to date on the integration of LLMs into IoT ecosystems. We cover various technical layers, including devices, software engineering, sensing, networking, data processing, privacy and security, and human interaction. We synthesize insights from more than 300 articles, identify open research gaps, and highlight promising future directions, ranging from semantic reasoning and communication to the integration of quantum computing. By addressing the characteristics, challenges, and opportunities associated with LLM integration, this article serves as a foundational resource for researchers and practitioners aiming to enhance IoT functionalities with LLMs while navigating the associated complexities.

INDEX TERMS Large language models (LLMs), Internet of Things (IoT), artificial intelligence, AI, survey.

I. INTRODUCTION

THE rapid proliferation of the Internet of Things (IoT) has led to the emergence of highly interconnected systems that integrate a wide range of devices. These systems provide significant benefits across various sectors, including environmental monitoring, healthcare, agriculture, and smart living environments [1]. Current estimates suggest that the number of connected IoT devices has surpassed 18 billion, with forecasts predicting 37.1 billion devices by 2032 [2]. Managing the vast network of IoT devices and their data is increasingly complex, especially in large-scale ecosystems. Conventional computing methods often struggle to cope with the diversity of IoT devices, the sheer volume of data, and the real-time demands of these environments. Consequently, there is a pressing need for advanced approaches capable of efficiently managing massive volumes of heterogeneous IoT data while facilitating intelligent and context-aware interactions.

Large Language Models (LLMs), recognized for their advanced natural language processing capabilities [3], have emerged as powerful tools for facilitating natural and context-aware interactions. These sophisticated neural networks, pre-trained on extensive datasets, can understand and generate human-like language. Their proficiency in processing large datasets allows tasks such as text summarization, analysis, and generation. These capabilities open up opportunities for meaningful conversations, personalized recommendations, and enhanced decision-making [4].

The integration of LLMs has the potential to enhance IoT ecosystems by managing device diversity, processing generated data, and addressing challenges within IoT environments. By leveraging LLM capabilities, IoT systems can achieve higher levels of intelligence and autonomy, leading to more effective operations. For instance, LLMs facilitate natural human-machine interactions [5], improve data interpretation through contextual understanding [6], [7], and

automate tasks [8], [9]. Commercial IoT applications harnessing LLMs are also increasingly emerging, such as BMW integrating with Amazon's Alexa to provide natural language assistance during driving [10], and PepsiCo's Frito-Lay factories utilizing LLMs for predictive maintenance to identify possible faults and reduce downtimes [11]. Another example is Microsoft's Azure IoT platform, which supports LLM inference and deployment for industrial IoT applications [12].

However, the integration of LLMs presents both opportunities and significant challenges. These challenges include the resource limitations of IoT devices, privacy and security concerns related to sharing data with LLMs, and the unique characteristics and interactions among different devices in IoT environments. Recent advancements have begun to address some of these challenges through techniques such as model optimization [13], edge LLMs [14], instruction tuning [15], and prompting strategies [16]. While these developments enable the utilization of LLMs in a broader range of IoT scenarios, the overall research landscape regarding the effective integration of LLMs into IoT systems remains in its nascent stage and is insufficiently explored. There is a lack of information, among others, about specific methodologies for seamless integration, scalability, and the potential security and privacy implications of deploying LLMs across diverse IoT environments.

This article rigorously surveys the research landscape at the intersection of LLMs and IoT, covering key challenges, current uses across the different functional layers of the IoT ecosystem, future and emerging trends, and open gaps. Existing surveys focus primarily on optimizing LLMs or integrating them into specific IoT components [17], [18], [19], [20], [21], [22], without providing a holistic overview of the entire IoT ecosystem. IoT applications are characterized by complex interactions across different functionalities. Without considering a holistic view of the entire ecosystem, the full potential of LLMs as well as many challenges and gaps may be overlooked. Our survey addresses this limitation in previous research by analyzing the challenges and opportunities associated with LLM integration into IoT systems. We review over 300 articles to synthesize advancements in LLM integration across IoT layers, from devices and software engineering to sensing, networking, and data processing. We also discuss security, privacy, and human interaction in LLM-powered IoT systems and their implications. Finally, we highlight future directions and emerging trends for research, as well as open research gaps identified from our findings. Addressing these gaps is crucial for maximizing LLM integration's potential and enhancing IoT capabilities. This survey serves as a foundational resource for researchers and practitioners looking to improve IoT functionalities with LLMs while navigating associated challenges.

II. SCOPE AND METHODOLOGY

Research on Large Language Models (LLMs) has surged exponentially in recent years. A Google Scholar search for "large language models" reveals 149 000

publications from 2023 to January 9, 2026, covering a wide range of topics, from foundational model architectures (e.g., GPT models [23], [24], LaMDA [25], LLaMA [26], BERT [27], and others) to innovative use cases across various application domains (e.g., healthcare [28], [29], [30], education [31], [32], environmental monitoring [33], [34], smart homes and cities [35], [36], and industrial IoT [37], [38]).

When narrowing the query to focus specifically on articles related to IoT, the volume of publications is smaller but shows a notably growing trend. Specifically, the query "Large Language Models" "LLMs" "Internet of Things" "IoT" results in more than 10 000 articles over the same period, with 530 results in 2023, 3000 in 2024, 6480 in 2025, and 171 as of January 9, 2026. Most of these studies focus on the research challenges of integrating LLMs with IoT technologies [39], or they provide specific examples of LLM applications within IoT contexts [36], [40]. However, a comprehensive understanding of the potential that LLMs can offer and the challenges they pose to IoT systems is notably lacking. Indeed, the integration of LLMs into IoT presents unique challenges and limitations, such as memory, energy, and other resource constraints, hardware heterogeneity, and privacy and security challenges. These factors complicate the deployment and use of LLMs as part of IoT ecosystems.

Our survey addresses this gap by providing a comprehensive and systematic review of LLMs across the entire IoT architecture. We emphasize both the breadth (covering all IoT layers) and depth (examining cross-layer interactions and challenges) of LLM integration, which distinguishes our work from prior surveys. This approach allows us to identify challenges and dependencies emerging from the integration of LLMs across all IoT layers, offering a holistic understanding of optimal deployments, resource allocation, and security and privacy considerations. Overall, our work bridges the existing knowledge gap and provides a foundational understanding of how LLMs can be effectively integrated into IoT frameworks.

A. RELATED SURVEYS

Table 1 provides a structured comparison of the most relevant surveys across several dimensions: LLM features, application domains, and coverage of the main IoT functional layers. Surveys focusing on *LLM features* delve into specific characteristics such as resource efficiency, architecture, training strategies, and prompting techniques. For instance, some reviews explore various prompting techniques for natural language processing (NLP) [16]. The use of LLMs has been extensively documented across numerous application domains, including healthcare [21], education [41], mental health [45], [48], personal assistance [19], and gaming [46]. However, these surveys address domain-specific issues and overlook IoT-specific challenges that arise from their integration into IoT applications.

In terms of IoT architecture, surveys on *IoT Devices* and *IoT Software Engineering* address the development of

TABLE 1. Comparison between related surveys and this survey, focusing on LLM background, domain, and IoT layers.

Reference	Year	Article corpus	LLM background				Domain				Devices		Software Engineering		Sensing		Networking			Data Processing			Privacy & Security		Human Interaction		Challenges / Limitations	Opportunities			
			Architecture	Prompt engineering	Fine-tuning	Reasoning	General	LLMs and IoT	Applications	Code generation	Task planning / management	On-device	Distributed	Development	Automation	Data generation	Context-aware sensing	Networking Technologies	Device Communication	Offloading	Analysis	Evaluation	Management	Privacy	Security	Intelligent UI			Personalization		
Bai et al. [17]	2024	✓	✓	✓	✓	✓			✓	✓	✓	✓	✓	✓	✓			✓	✓							✓	✓	✓			
Chang et al. [41]	2024	✓		✓	✓	✓	✓			✓										✓	✓					✓	✓	✓			
Cui et al. [42]	2024		✓	✓	✓		✓		✓													✓	✓				✓	✓			
Das et al. [43]	2025	✓		✓	✓				✓	✓												✓	✓				✓	✓			
Ferrara et al. [21]	2024		✓	✓	✓			✓	✓			✓			✓	✓			✓	✓		✓	✓				✓	✓			
Friha et al. [44]	2024	✓		✓		✓	✓			✓		✓	✓				✓	✓									✓	✓			
Guo et al. [45]	2024	✓			✓				✓						✓					✓			✓	✓				✓	✓		
Hu et al. [46]	2024		✓	✓			✓		✓	✓															✓	✓		✓	✓		
Jin et al. [47]	2023	✓		✓		✓								✓						✓					✓	✓			✓		
Kök et al. [39]	2024				✓			✓		✓										✓			✓	✓				✓	✓		
Lawrence et al. [48]	2024			✓	✓				✓						✓					✓			✓	✓				✓	✓		
Li et al. [19]	2024	✓			✓			✓								✓				✓		✓	✓	✓	✓	✓		✓	✓		
Minaee et al. [4]	2024		✓	✓	✓	✓	✓			✓																			✓	✓	
Naveed et al. [18]	2023		✓	✓	✓	✓		✓		✓		✓	✓			✓				✓	✓							✓	✓		
Qu et al. [49]	2025	✓	✓	✓	✓		✓		✓		✓	✓	✓					✓	✓				✓	✓				✓	✓		
Wang et al. [50]	2024	✓		✓	✓	✓		✓																				✓	✓		
Yao et al. [3]	2024	✓		✓	✓	✓			✓	✓			✓	✓					✓					✓	✓			✓	✓		
Zhang et al. [15]	2023	✓	✓	✓	✓		✓			✓						✓														✓	✓
Zhang et al. [22]	2024	✓	✓	✓				✓	✓							✓				✓								✓	✓		
Zhou et al. [51]	2024	✓	✓	✓	✓		✓		✓	✓	✓							✓	✓	✓				✓					✓	✓	
Zhu et al. [52]	2023	✓	✓	✓	✓	✓		✓												✓	✓							✓	✓	✓	
Our survey	2025	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Total of Surveys		15	12	19	18	5	16	6	13	11	5	7	5	3	2	7	4	3	6	5	10	5	2	13	13	3	5	19	16		

LLM-based agents and instruction tuning for GPT models, while those focused on *IoT Sensing* investigate LLMs for sensor data retrieval and model tuning. However, there is a notable scarcity of literature on *IoT Networking*, with current surveys primarily covering telecommunication networks or edge computing [39], [44], [49], [51]. Surveys on *IoT Data Processing* evaluate methods for analyzing sensor data and multimodal datasets using LLMs, and those addressing *IoT Privacy and Security* examine vulnerabilities associated with LLMs and their implications for privacy and security at both the system and user levels.

As synthesized from Table 1, the existing literature exhibits a thematic skew. While surveys thoroughly emphasize the architectural background of LLMs and individual application domains, they systematically underrepresent the core functional layers of IoT. Rather than examining the complexities of LLM-IoT integration and the challenges that arise for IoT deployments, current reviews treat IoT systems as endpoints that produce data for LLMs. As a result, critical areas such as task planning are largely overlooked, appearing in fewer than 30% of surveys. Nearly 40% of surveys offer no substantive treatment of core layers, including networking, sensing, software engineering, or devices, and only one survey addresses all the mentioned layers (7 out of 9 areas). Data processing coverage is similarly imbalanced. Indeed, data analysis is discussed in nine surveys (42%), but data management appears only once and evaluation in four. Human interaction is notably underrepresented, covered by four surveys (19%) while intelligent UIs are discussed only in two (9.5%). Furthermore, while security and privacy are discussed in the majority of surveys, 38% of surveys omit these topics entirely. Challenges and opportunities are typically acknowledged but

vary in depth. Most surveys include some discussion, and only two (9.5%) provide no coverage. While these surveys have explored specific LLM characteristics or isolated IoT layers, they overlook the significant interactions across the entire IoT landscape. A comprehensive understanding of these interactions is essential for effectively harnessing LLMs in IoT applications, as many fundamental architectural layers and key components such as hardware and software, sensing and data analysis, and communications and networking, remain largely unexplored.

As this comparison indicates, existing surveys primarily address isolated challenges or application domains within individual IoT domains or layers, lacking a holistic view of the overall IoT ecosystem. This significantly restricts the insights they can derive about interactions between different IoT components and subsystems. Our survey addresses this critical gap by systematically investigating the potential of LLMs across all IoT layers and identifying open challenges both within and between these layers. By offering a holistic perspective on the integration of LLMs into various IoT layers, this paper provides a comprehensive understanding of their applications, potential benefits, limitations, and the challenges associated with utilizing LLMs in the IoT landscape.

B. ANALYTICAL FRAMEWORK AND ARTICLE STRUCTURE

We conduct a rigorous analysis of the open research challenges in the field and provide insights into how LLMs can be tailored to meet the specific demands of IoT systems. Our analysis is grounded in a three-pillar framework that defines the technical viability of LLM-IoT integration: (i) **Resource Efficiency** measures the trade-off between model intelligence

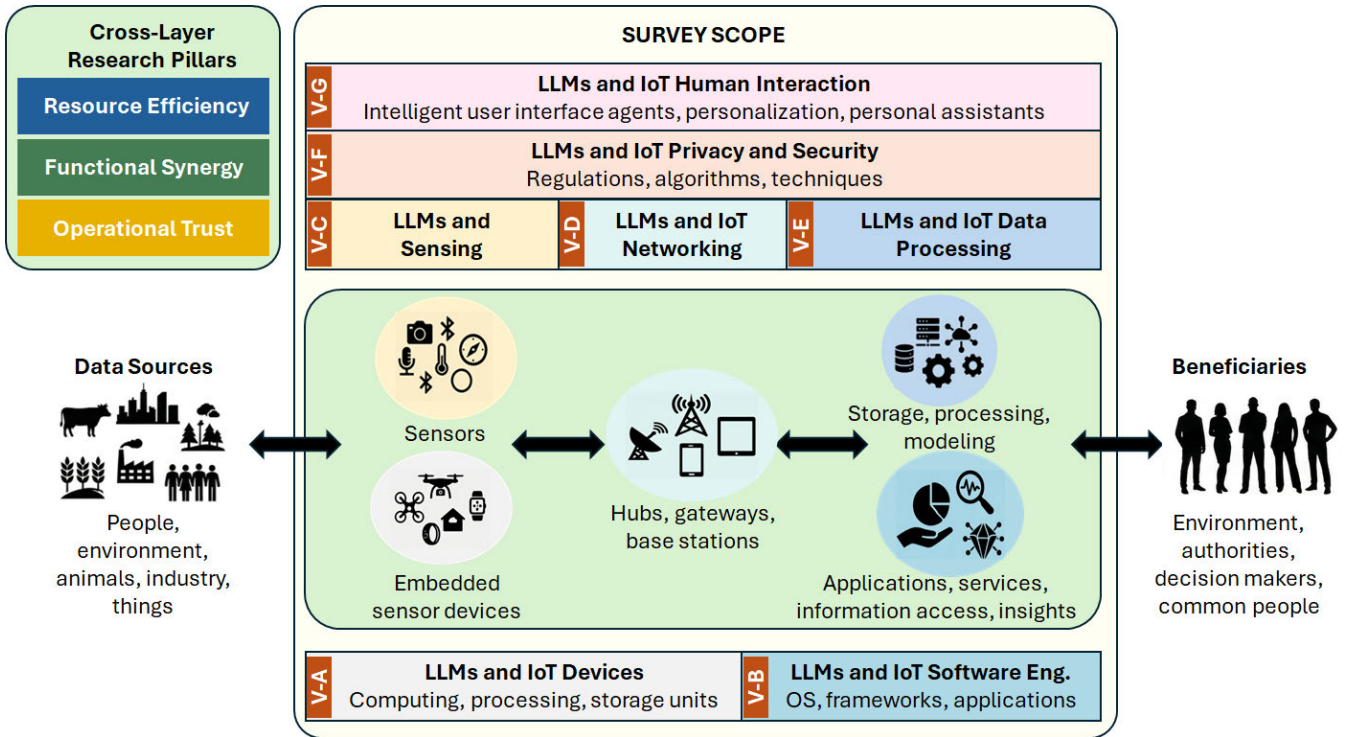


FIGURE 1. LLMs and their Interaction with Functional Layers of IoT Applications. Sensing, networking, and data processing are classified as specialized functional layers, each addressing distinct aspects of IoT operations. In contrast, devices, software engineering (Eng.), privacy and security, and human interaction are cross-cutting concerns that influence the entire IoT ecosystem, affecting all systems and deployments. Our survey addresses LLMs integration within all functional layers, evaluating each through our three-dimensional framework: resource efficiency, functional synergy, and operational trust. (Note: The icons used in this figure were generated using Google Gemini).

and the physical constraints of IoT hardware, such as power, memory, and bandwidth; (ii) **Functional Synergy** evaluates the degree of interoperability and collaborative reasoning achieved between LLM agents and deterministic IoT protocols; and (iii) **Operational Trust** assesses the systemic reliability, security, and privacy of the resulting LLM-driven IoT ecosystem. These pillars establish the criteria for assessing how effectively LLMs meet the functional requirements of modern IoT deployments.

We start with an overview of current LLM technologies in Section III and briefly cover the challenges associated with the deployment of LLMs in Section IV. The core of the survey, presented in Section V, is structured to cover the primary technological layers of IoT, including devices, software engineering, sensing, networking, data processing, privacy and security, and human interaction. An overview of the overall scope of the survey is shown in Figure 1. Section VI synthesizes our analysis, presenting future directions and emerging trends as well as identifying open gaps for future research. Finally, Section VII concludes the article.

C. LITERATURE SELECTION PROCESS

The article selection and analysis process for this survey was conducted in three stages: article identification, preliminary screening, and comprehensive article analysis. Each stage was designed to systematically identify the most relevant and best quality articles to include in the final survey.

1) STAGE 1: ARTICLE IDENTIFICATION

In this first stage, we conducted a systematic search across three major academic databases: ACM Digital Library, IEEE Xplore, and Google Scholar. This search was first performed in September 2024, and was repeated on February 15, 2025. We employed keywords and their possible combinations of *Large Language Models*, *LLMs*, *Internet of Things* and *IoT*. This approach ensured that we captured articles discussing LLMs in the context of IoT from different perspectives. We supplemented the results by searching for follow-up works and articles that cite, and are cited by them. Duplicate versions of the same article (e.g., preprints, published versions, or conference presentations) were removed.

2) STAGE 2: PRELIMINARY SCREENING

In the second stage, we evaluated the articles identified in the first stage. The preliminary screening involved reviewing abstracts to assess their relevance across several key aspects: (i) LLM architectures: technical details, datasets used for model training, and interfaces, and (ii) the utilization of LLMs in IoT: knowledge acquisition, feature engineering, inference, recommendation, privacy, security, and user interaction. Each article was then tagged as *potentially relevant* or *irrelevant* based on its alignment with the survey's scope and these key aspects. Accordingly, we excluded articles that merely mentioned LLMs in passing or as part of broader discussions without a focused analysis. Additionally, articles that focus

solely on theoretical aspects of LLMs or IoT without practical applications were excluded. This criterion was essential to keep our survey focused on research providing insights and solutions to real-world challenges. To finalize the selection of articles, we employed a majority decision-making process among the authors. This collaborative approach ensured that the final selection reflects a consensus on the relevance and quality of the articles, allowing us to compile a comprehensive and insightful survey. Our final corpus includes over 300 articles. Links to these articles, organized by the functional IoT layer they address, are available in a separate GitHub repository.¹

3) STAGE 3: COMPREHENSIVE ARTICLE ANALYSIS

In the final stage, we conducted a comprehensive analysis of the articles in the corpus. We meticulously reviewed each article to identify and label key concepts (e.g., feature extraction, preprocessing time-series data) and phrases (e.g., generating IoT application scenarios) relevant to our survey scope. This step allowed us to capture a wide range of information without imposing predefined categories, fostering a broader understanding of the articles on how IoT systems can leverage the capabilities of LLMs. We then organized the extracted information into categories and explored the relationships among them to uncover insights into the complexities and nuances of LLM integration with IoT systems. Our analysis systematically examines both the breadth and depth of the literature. This provides a comprehensive understanding of the current LLM-IoT landscape. Furthermore, it identifies key research gaps to guide future investigations and practical applications. Lastly, building on this foundational body of literature, we identified key ideas, research directions, and emerging technologies that can further advance the integration of LLMs and IoT. This broader perspective highlights how their convergence is driving innovation across multiple fields. Taken together, our survey serves as a foundational resource for researchers and practitioners in these fields, equipping them with essential insights to drive innovation to address existing challenges and explore new avenues for effective integration of LLMs and IoT.

III. LARGE LANGUAGE MODELS

Large Language Models (LLMs) are advanced neural networks that have been pre-trained on extensive datasets containing diverse forms of data, including text, code, and more. This pre-training equips them with the ability to (i) deeply understand and extract information from human-like text, (ii) generate coherent and contextually relevant text, (iii) infer context and demonstrate contextual awareness, and (iv) provide recommendations or instructions [3]. Notable examples of LLMs include the Generative Pre-trained Transformer (GPT) series, BERT [27], XLNet [53], ChatGLM [54], LLaMA [26], PaLM [55], Claude [56], DeepSeek [57], Gemini [58], and Grok-3 [59].

We refer to [60] for an up-to-date list of LLM tools and their performance in different tasks. There have also been efforts to create specialized models that address specific aspects of IoT or target IoT-specific industries. For example, Google SensorLM is a research project focusing on creating a foundation model for wearable data [61]. Examples of enterprise LLMs, in turn, include SymphonyAI [62], H2O.ai [63], [64], Marble [65], and research initiatives by Microsoft to establish industry-specific foundation models [66].

The development of LLMs is rooted in the evolution of language modeling (LM) techniques. Initially, language modeling served as a foundational approach for understanding and manipulating human language within the field of natural language processing (NLP). Early models utilized statistical methods to calculate the probability distributions of word sequences [67]. Over time, these models evolved into statistical language models that predict the next word in a sequence. With the advent of deep learning, SLMs transitioned into neural language models (NLMs), which emphasize the interaction between human and computer language [68]. Pre-trained language models (PLMs) represent a specific category of language models that leverage large datasets to transfer knowledge from previously learned tasks to new applications [69]. PLMs with substantial amounts of data and a large number of parameters (often exceeding 6 billion) are commonly referred to as Large Language Models (LLMs) [68]. This evolution not only improved the performance of language models but also broadened their applicability across various domains, marking a significant milestone in the field of NLP. More recently, Large Action Models (LAMs) are special variants of LLMs that extend LLM capabilities to deliver agents that autonomously perform real-world actions [70].

At the core of LLMs lies the transformer neural network architecture, which operates on *tokens*. A token is a unit of input data that can be transformed into a vector representation, serving as the processing unit within the transformer model. The process of converting input data into tokens is referred to as *tokenization*, where the tokens can represent characters, words, or other elements. This tokenization enables efficient processing and analysis of text. The transformer architecture employs two key mechanisms that facilitate the learning of dependencies and relationships between tokens. The first mechanism is *self-attention*, which allows the model to weigh the importance of each token relative to others, capturing internal correlations and contextual relationships. The second mechanism, *multi-head attention*, enables the model to perform multiple self-attention operations in parallel, with each “head” focusing on different aspects of the input. By capturing a broader range of dependencies, multi-head attention enhances the model’s ability to understand complex relationships within the data [71].

The transformer model itself consists of two main components: the *encoder* and the *decoder*. The encoder processes the input data, extracts features, and learns the relationships between tokens using self-attention mechanisms. It embeds

¹<https://github.com/PDSUH/llms-iot>

contextual information along with the input sequence into a series of vectors for the decoder to utilize. The decoder generates output data sequentially based on the input received from the encoder. In addition to multi-head self-attention and feed-forward networks, the decoder incorporates encoder-decoder attention, which connects it with the encoder. This connection enables the decoder to compute attention scores for all tokens in the encoder's output, determining which parts of the input sequence are relevant for generating the output.

During a query, each token is transformed into three vectors: a Query vector, a Key vector, and a Value vector. The Query vector contains information of interest, the Key vector represents each token's unique characteristics, and the Value vector holds the actual information associated with each token. By comparing the Query vectors with the Key vectors, the model computes attention scores that indicate the relevance of each token in relation to others. Throughout this process, self-attention is iteratively applied across layers, capturing dependencies that are stored in memory as attention key-value tensors, collectively referred to as the key-value (KV) cache [72]. The storage and retrieval of the KV cache and model parameters from memory scale the inference cost quadratically with input sequence length [73]. In fact, the KV cache for a single request can be as large as 1.6 GB [74]. Insufficient management of the KV cache may exhaust memory that could otherwise be allocated for processing incoming requests.

To mitigate the memory footprint associated with LLMs, the KV cache can be offloaded to CPU memory, either on the same device or to another device over the network [75]. Additionally, previously computed key/value vectors can be reused for generating the current token through techniques such as KV cache compression [76]. However, offloading the KV cache, whether locally or across a network, can introduce latency, which may hinder performance. This challenge is especially pronounced in resource-constrained environments, such as Internet of Things (IoT) devices, where limited CPU memory complicates the management of large caches effectively. Consequently, there is a pressing need for innovative memory management strategies to ensure efficient deployment of LLMs in such settings.

Traditional LLMs utilize both an encoder and a decoder to process input and generate output. However, the demands for efficiency, simplicity, and task specialization have led to the rise of encoder-only and decoder-only models, each optimized for specific applications. Encoder-only LLMs, such as BERT [27] and ALBERT [77], utilize the bidirectional self-attention mechanism of the encoder component of the transformer architecture. This allows them to capture contextual information and dependencies from the input data effectively, resulting in rich contextual embeddings that represent the semantics of the input. Encoder-only models excel in tasks that require a deep comprehension of the input data, making them efficient for downstream tasks such as sentiment analysis, feature extraction, and classifica-

tion [78]. In contrast, decoder-only LLMs, such as GPT [24], Claude [56], and LLaMA [26], utilize an auto-regressive self-attention mechanism within the decoder component of the transformer architecture. This design enables decoder-only models to generate text sequentially based on previously predicted tokens to maintain coherence and context. This makes decoder-only models particularly well-suited for tasks that involve creating coherent and contextually relevant sequences of text, such as conversational agents, contextual reasoning, creative writing, code generation, and summarization [78]. These architectural variations offer flexibility to tailor LLMs to specific task requirements. Tasks focusing on understanding and analyzing data without the need to generate responses are best served by encoder-only models, whereas tasks that necessitate real-time response generation without the need for understanding or analyzing the data benefit from decoder-only models. Accordingly, tasks that require both capabilities should utilize models that integrate both the encoder and decoder components.

IV. CHALLENGES IN LLM AND IoT INTEGRATION

The integration of LLMs with IoT systems poses significant challenges due to the complexities of both technologies. We next discuss the key challenges in the use of LLMs, applicable across the entire IoT ecosystem. Issues related to individual IoT layers are discussed in the next section. While these challenges are not exclusive to IoT, understanding them is crucial for gaining insights into the research landscape and ensuring the effective integration of LLMs and IoT technologies.

A. RESOURCE CONSTRAINTS AND SUSTAINABILITY

LLMs require immense computational power, memory, and energy, while many IoT devices have limited resources. This can lead to inference latency, high power consumption, and challenges in storing LLM parameters. Techniques to optimize the efficiency of LLMs include model optimization techniques discussed below, and hardware acceleration as discussed in Section V-A. Besides the model itself, the key-value cache used by LLMs also demands significant memory, exacerbating resource demands. Strategies to mitigate memory usage include efficient compression techniques and strategic cache management [79], [80]. Many IoT devices are battery-powered, and the high energy requirements of LLMs pose sustainability concerns [81]. Optimizing resource utilization is critical for operational efficiency and sustainability in LLM applications within IoT environments.

B. MODEL OPTIMIZATION

Model optimization methods focus on reducing LLM model size and resource demands, but often do so at the cost of accuracy. Key methods include pruning, quantization, and knowledge distillation [82], [83]. Pruning eliminates less significant parameters, layers, or channels, making it well-suited for edge and on-device deployments [84], [85],

[86]. IoT-specific pruning methods, such as DeepIoT [87] and FastDeepIoT [88], focus specifically on achieving better compression-accuracy trade-offs while ensuring compatibility with resource-constrained devices. Quantization reduces resource demands by adjusting parameter precision, which is particularly effective on low-end devices [13], [89], [90]. Modern quantization often employs mixed-precision modeling, utilizing 16-bit and 32-bit representations for different parts of the model [91]. While this accelerates inference and training, it may lead to instability, resulting in loss spikes due to inaccuracies in gradient estimates. Knowledge distillation transfers knowledge from larger models to smaller ones [92], though it can reduce model diversity and adaptability [93], [94]. The DeepSeek model family exemplifies how these techniques can lower inference requirements for conventional hardware [57], [95]. Another option for optimizing model efficiency is to use neural architecture search (NAS) techniques to identify a smaller model architecture that closely resembles the original while meeting the resource constraints of the target platform, with promising results reported in applying NAS to LLMs [96]. However, a major drawback of these techniques is that finding the optimal model architecture often requires significant computational resources, which can limit the ability to adapt models on demand.

C. PROMPT ENGINEERING

Prompt engineering involves crafting rules and instructions to maximize LLM capabilities based on specific data types. Common methods include zero-shot prompting, which simplifies interactions by using an LLM without a detailed prompt, and zero-shot learning, which enables models to predict unseen classes using pre-learned attributes [97], [98]. This approach has been effectively applied in IoT domains, such as healthcare and smart homes, by incorporating contextual information into prompts [99], [100], [101], [102]. Generalized zero-shot prompting enhances adaptability by capturing semantic relationships, while few-shot learning improves performance by leveraging a small number of labeled examples, often outperforming zero-shot prompting [100], [103]. Advanced techniques, such as Chain of Thought (CoT) prompting, provide step-by-step explanations to improve reasoning quality [104]. CoT has been successfully applied in activity recognition and task planning in smart homes [36], [105], while self-consistency (SC) prompting enhances logical consistency, especially in complex fields like healthcare [100]. Other methods, like Program of Thought (PoT) prompting, enable LLMs to generate code for complex tasks [106]. Recently, token-budget-aware methods aim to compress LLMs' reasoning process by considering the token budget for each prompt, for example, Han et al. [107] predict an adjustable token budget for each CoT prompt. Similarly, SelfBudgeter [108] utilizes reinforcement learning to predict the token budget based on problem complexity, and Liu et al. [109] make agents budget-aware

using a budget tracker to optimize resource usage. While these methods offer efficient LLM reasoning, they are limited to text data. Despite these advancements, IoT systems face challenges due to diverse environments and data types, which can lead to large, computationally expensive prompts that reduce responsiveness. Thus, optimizing prompts to balance the accuracy of reasoning with computational efficiency specifically designed for IoT environments is essential for seamless LLM integration in resource-constrained environments.

D. AGENTIC LLMs

Agentic LLMs extend conventional prompting by enabling planning, tool use, memory, and autonomous decision-making [110]. In IoT systems, Agentic LLMs act as orchestrators over heterogeneous devices and services, discovering capabilities, composing APIs, and executing multistep tasks under resource constraints. Multi-Agent LLM systems distribute roles across specialized agents with domain-specific tools and policies to improve reliability and scalability. Agents can adapt dynamically to various scenarios, analyze multimodal inputs, generate strategies and reason under uncertainty, and engage in self-improvement [111], [112]. Agentic AI and multi-agent LLMs transform LLMs into autonomous orchestrators capable of planning, tool use, and coordinated control. Agentic AI has been used in various applications such as automated prescriptive maintenance in smart manufacturing [112], distributed IoT automation system (LLMind 2.0) [113], spatiotemporal urban data analysis (CityGPT) [114], and industrial IoT system enhancement (LUMEN) [115]. Moreover, using SLMs on edge devices further improves Agentic LLMs by preserving privacy and reducing latency through local deployment [111].

E. FINE-TUNING

Fine-tuning LLMs on task-specific datasets can significantly enhance response time and accuracy. Fine-tuning is also the most common method for adapting LLMs to enterprise and IoT settings [116]. One effective approach is integrating explicit instructions as training examples, which is particularly beneficial in domains with clear procedures, such as healthcare [100], [103]. The self-instruct technique [117] automates instruction tuning using minimal human-annotated data. Parameter-efficient fine-tuning (PEFT) methods, like Adapter Tuning [118] and Low-Rank Adaptation (LoRA) [119], adjust a small subset of parameters, reducing time and computational resources compared to full fine-tuning [120]. While Adapter Tuning may slightly increase latency, LoRA does not. PEFT methods have been applied to various IoT tasks, including synthesizing RF data [121], developing mobile firmware [122], and predicting traffic patterns [123]. Other approaches, such as LoFit [124], aim to enhance interpretability and truthfulness. Test-time scaling is a recent technique that combines reinforcement learning with supervised fine-tuning to improve domain-specific reasoning in

pre-trained LLMs [125]. This method uses a small set of carefully selected samples and limits the number of test tokens generated, allowing LLMs to verify and correct their reasoning steps [125]. It can also leverage different architectures to optimize resource use, such as employing depth-recurrent blocks in transformers for efficient computation without requiring special training data, thereby improving reasoning capabilities [126].

F. RETRIEVAL-AUGMENTED GENERATION (RAG)

Retrieval-Augmented Generation (RAG) is a promising approach to customize LLMs for specific domains. RAG integrates external knowledge retrieval with generative capabilities, allowing LLMs to combine relevant, up-to-date, and context-specific information without extensive retraining or dedicated fine-tuning, resulting in more accurate responses [127], [128]. RAG can also reduce the computational burden on IoT devices by offloading knowledge retrieval to more capable edge or cloud resources, making RAG a key enabler for scalable and intelligent IoT applications. This approach has been applied in various IoT domains, including real-time environmental monitoring [127], secure data management [129], chat-based security assistants [130], and automated test-case generation [130]. Various optimized RAG techniques have been developed to enhance energy and memory efficiency for mobile devices [128], [131].

G. MODEL EVALUATION

Evaluating LLMs presents unique challenges, particularly for IoT-specific tasks. Traditionally, LLM evaluation focused on text-based metrics, but recent efforts have shifted towards benchmarks tailored for IoT applications. For instance, eAIBench [122] covers various mobile AI tasks across multiple data modalities, including images, voice, text, and sensor data, specifically designed to assess LLM-based mobile firmware. Another benchmark, SensorQA [132], features human-generated question-answering datasets centered on sensor data, providing rich evaluation context. Recent benchmarks on SensorQA reveal a significant performance gap in multi-timescale reasoning. While state-of-the-art models can identify simple daily activities with high accuracy, their performance on mobile sensor QA can be as low as 31.22% for certain configurations. Models frequently struggle with “temporal aggregation” tasks, such as reasoning over extended periods like multiple weeks, often due to the context window bottleneck and the noise inherent in real-world sensor streams [132]. Beyond output accuracy, there is increasing emphasis on assessing LLM reasoning capabilities, including understanding causal relationships and drawing inferences. The Order-of-Thought framework evaluates reasoning skills based on sensor data by incorporating varying levels of guidance into zero-shot, chain-of-thought (CoT), and prompt-of-thought (PoT) strategies [106]. Fine-tuned LLMs can also serve as evaluators for other models through the LLM-as-a-judge approach, where they are trained

on datasets with specific instructions, desired responses, and evaluation criteria [133]. Fine-tuned LLMs can generate prompts and assess responses from other models, providing a structured evaluation mechanism. Alternatively, general-purpose LLMs like GPT-4 can be used for evaluation due to their proficiency in following instructions. By offering a prompt template that includes both the instruction and expected response, these models can effectively evaluate other LLMs’ performance in the LLM-as-a-judge approach. This helps ensure LLM robustness and alignment with the requirements of IoT applications. TinyLLM [134] provides evaluation and optimization for agentic small language models (SLMs) used as agents on edge devices. Evaluation results within the TinyLLM framework show that medium-sized SLMs (1–3B parameters) significantly outperform ultra-compact models (<1B parameters), with top-performing models like xLAM-2-3b achieving up to 65.74% overall accuracy in tool-use tasks. However, in complex multi-turn agentic workflows, critical for real-world IoT interactions performance remains a challenge, with multi-turn accuracy for these same models dropping to 55.62%. These results underscore that while SLMs provide efficient local deployment, maintaining logical consistency during long-running autonomous tasks remains an open research gap [134].

H. TRUTHFULNESS

A significant challenge in integrating LLMs into IoT systems is hallucination, where LLMs generate false information. This is particularly dangerous in safety-critical domains like healthcare and autonomous systems, as it compromises reliability [42]. Ensuring the accuracy of LLM outputs is crucial for effective decision-making in IoT environments. Truthfulness can be enhanced, for example, by incorporating a bias vector into specific attention layers [124] or by using contrast decoding to eliminate false information [135]. Additionally, lie and hallucination detection methods are vital for maintaining the integrity of LLM-generated data [136], [137]. Identifying hidden states that encode truth representations can further enhance truthfulness [138], and this technique can also be extended for hallucination detection. For instance, Zhang et al [137] employed prompting techniques to control LLM internal states based on text truthfulness. Another solution is collaborative LLMs, where multiple models cross-validate outputs by referencing each other, refining inferences through peer review and self-tuning mechanisms [139], [140].

I. TRUSTWORTHINESS AND ETHICS

Trustworthiness is a critical concern, especially given the closed-box nature of LLMs, which complicates understanding their inference processes [141]. It is an integral part of ethical use, along with other essential qualities for responsible technology deployment, including interpretability, reliability, explainability, safety, fairness, transparency,

and resilience [44], [142], as outlined in regulatory frameworks like the EU AI Act and the US AI Act [143]. In IoT systems, fostering trustworthy interactions is paramount, as these devices often cater to individual preferences and require fair and transparent inference capabilities. To enhance trust, integrating clear reasoning instructions into LLM prompts can elucidate the decision-making process, making the underlying data and algorithms more transparent. Additionally, LLM decisions should be summarized and stored for a defined period or presented to end-users in a digestible format [142]. Solutions like audio explanations can further improve accessibility and user experience while promoting the responsible integration of LLMs in IoT systems.

J. PRIVACY AND SECURITY

LLM integration into IoT ecosystems presents significant privacy and security challenges, alongside unique opportunities for enhanced protection [144]. IoT devices routinely collect and transmit sensitive data, such as health information and location details, which can be further exposed when processed by LLMs, especially if this data is sent to the cloud or third-party services [3], [144]. This data flow increases the risk of breaches, unauthorized access, and inadvertent leaks, particularly given the vast datasets required to train LLMs, which may inadvertently contain or replicate private user information [145], [146]. Furthermore, LLMs introduce new attack surfaces. For example, adversaries can exploit vulnerabilities through data poisoning, adversarial prompts, or model inversion attacks to manipulate device behavior or extract confidential data [147], [148], [149]. The resource constraints of many IoT devices complicate the deployment of robust protections, as advanced security mechanisms like encryption and anomaly detection may be challenging to implement at scale or in real time [39]. Additionally, employing federated learning or distributed LLM inference introduces further risks, including model poisoning and gradient manipulation [150], [151]. While LLMs can enhance security by enabling intelligent threat detection and adaptive policy enforcement, their outputs are susceptible to biases and hallucinations, which may undermine trust and reliability [152], [153]. As LLM-driven IoT applications continue to proliferate, it is essential to develop context-aware, resource-efficient, and resilient privacy and security strategies that effectively address both the capabilities and limitations of these technologies.

Practical integration challenges in real-world scenarios. Despite active progress in optimization and prompting, field deployment exposes constraints that are not fully captured by lab benchmarks. First, energy and latency budgets on battery-powered devices limit model size, context windows, and multi-turn agent workflows. While offloading reduces local load, it introduces latency under variable networks. Second, continuous sensing and multimodal fusion increase privacy risks and require PETs and TEEs that add overhead and engineering complexity. Third, KV-cache management,

memory fragmentation, and accelerator heterogeneity complicate portable inference across fleets. Fourth, evaluation on realistic workloads is scarce. Systems should report end-to-end metrics (task completion, user-perceived latency, energy per task, privacy leakage, and robustness to drift) rather than isolated model accuracy.

K. CONCLUDING REMARKS

The challenges of integrating LLMs with IoT are summarized in Table 2. Despite advancements in techniques like prompt engineering, fine-tuning, and knowledge distillation, their effectiveness in real-world IoT applications is context-dependent. For instance, while few-shot prompting can enhance performance, it may introduce latency, making it unsuitable for real-time use. Knowledge distillation can reduce resource demands but may compromise accuracy and adaptability. In safety-critical IoT systems, transparent and reliable decision-making is essential, highlighting the need for ethical considerations and privacy protection. Ultimately, successful deployment of LLM techniques requires balancing resource efficiency, reliability, accuracy, and trustworthiness according to the specific needs of each application. Understanding these challenges and exploring solutions is crucial for effective integration.

V. LLMS AND INTERNET OF THINGS

This survey provides a comprehensive review of the integration of LLMs with IoT across all functional layers of the IoT ecosystem, serving as a reference for researchers and practitioners. We now move to the main part of our survey, structuring the material following different IoT layers. The topics we discuss along with examples of papers addressing them are summarized in Table 3.

A. IoT DEVICES

LLMs can increasingly operate on IoT devices even if most of them have not been originally built for this purpose. We next examine ways to improve the integration of LLMs with IoT devices, evaluating these approaches through our analytical framework. We cover on-device models, distributed architectures, and hardware accelerators.

1) ON-DEVICE LLMs

Driving improvements in resource efficiency, on-device LLMs are highly optimized versions of large language models specifically designed for embedded devices, such as mobile firmware [122] and LLM-based operating systems for automated vehicles [9]. These models leverage model compression techniques introduced in the previous section, such as pruning and quantization, to maximize resource efficiency within the constraints of embedded environments [14]. Examples of on-device LLMs that utilize quantization methods include TinyAgent [90] and LLMEdge [13], and more recently Microsoft Phi-3 and Phi-4-mini [160], Gemma3-4B-IT [161], and Qwen 2.5 [162]. Recent advancements, such as Huff-LLM, introduce innovative lossless

TABLE 2. Summary of LLM challenges, their relevance to IoT, and their applications within IoT.

Topic	Relevance to IoT	Applications within IoT
Resource Constraints and Sustainability	LLMs require significant computational resources, which many IoT devices lack. High energy consumption poses sustainability challenges.	Optimizing resource utilization is essential for operational efficiency in battery-powered IoT devices. Techniques like model compression and hardware acceleration are crucial.
Model Optimization	Reduces the size and resource demands of LLMs, making them feasible for resource-constrained IoT devices.	Techniques like pruning, quantization, knowledge distillation, and subnetwork modeling applied in edge and on-device deployments, improving performance in IoT applications.
Prompt Engineering	Enhances LLM capabilities tailored to specific data types and contexts.	In healthcare and smart homes to improve LLMs interaction, incorporating contextual information and optimizing for computational efficiency.
Fine-tuning	Adapts LLMs to specific tasks to improve response time and accuracy.	IoT tasks like synthesizing RF data, developing mobile firmware, and predicting traffic patterns, enhancing domain-specific LLMs use cases.
Retrieval-Augmented Generation (RAG)	Integrates knowledge retrieval with LLM capabilities, reducing computational load.	Used in real-time environmental monitoring, secure data management, and automated test-case generation, enhancing accuracy and efficiency.
Model Evaluation	Addresses unique challenges in assessing LLM performance for IoT-specific tasks.	Utilizes benchmarks tailored for IoT applications to evaluate reasoning capabilities and ensure robustness in IoT settings.
Truthfulness	Ensures LLM outputs accuracy, which is critical for decision-making in safety-critical systems.	Techniques for enhancing truthfulness and detecting hallucinations, ensuring data integrity in healthcare and autonomous systems.
Trustworthiness and Ethics	Addresses the ethical implications of using LLMs in IoT, ensuring fair and transparent interactions.	Integrating clear reasoning in LLM prompts and summarizing decisions enhances user trust and promotes responsible technology deployment.
Privacy and Security	Highlights the privacy and security challenges of integrating LLMs with IoT, including data breaches and adversarial attacks.	Advocates for context-aware security measures, such as encryption and anomaly detection, to protect sensitive data and enhance resilience against cyber threats.

compression of LLM weights based on Huffman coding, along with a customized decoder, to significantly reduce

model size and energy consumption [154]. Additionally, some works harnessing knowledge distillation have been proposed [155], [156]. For instance, MiniLM [157] employs deep self-attention layers to accurately emulate the behavior of larger LLMs with a reduced parameter count. Ray [158] identifies a specific class of sub-billion parameter models, such as SmolLM (135M/360M) and Qwen-0.5B, which are capable of running on constrained edge hardware. Additionally, Chen et al. emphasize that PEFT methods like LoRA reduce the memory required for on-device adaptation by over 70% compared to full fine-tuning [159]. Similarly, the Huff-LLM framework demonstrated a significant reduction in energy consumption (up to 40%) for edge devices by using lossless compression. Future directions to enhance functional synergy include the integration of on-device LLMs with federation and other distributed learning paradigms to enable collaborative model training. Hardware advancements, such as neuromorphic architectures, may also be able to further improve resource efficiency and enable more sophisticated LLM capabilities on constrained devices.

2) DISTRIBUTED LLMs

To establish functional synergy across the network, resource requirements can be minimized by collaboratively distributing LLMs across edge and terminal devices [163]. This strategy facilitates the cooperative optimization of performance trade-offs while accounting for available resources. In [164], a collaborative fine-tuning framework is proposed that utilizes a backbone LLM and adapter layers on edge nodes to adapt models according to their specific resource constraints. CoLLM [165] presents a cooperative deployment strategy that integrates tensor parallelization, adaptive load balancing, and an algorithm to determine the optimal number of collaborators for minimizing latency. Similarly, Edge-LLM is designed to optimize both computational and memory requirements for edge devices [86]. To achieve this, Edge-LLM employs a layer-wise compression method along with an adaptive layer tuning method that uses voting to select the best output. Furthermore, Edge-LLM includes a hardware scheduling module to streamline irregular computation patterns arising from diverse quantization, compression, sparsity, and updates to LLM segments. Despite these advancements in functional synergy, resource constraints remain a significant challenge for deploying distributed LLMs in IoT applications. Future directions involve the development of algorithms and optimization frameworks for managing resource usage, and tighter integration with fast-speed communication technologies to improve communication efficiency between LLMs.

3) DYNAMIC SUBNETWORK SELECTION

Dynamic subnetwork selection is a specialized optimization technique that leverages the structure of LLMs to activate only specific subsets of the model, known as *subnetworks*. These subnetworks can be distributed across different devices to optimize resource efficiency and resource utilization [71].

TABLE 3. LLMs in IoT layers.

IoT Layer	Topics	References
LLMs and IoT Devices	On-Device LLMs	[9], [13], [14], [90], [122], [154]–[162]
	Distributed LLMs	[86], [163]–[165]
	Dynamic Subnetwork Selection	[166]–[171]
	Hardware Accelerators	[172]–[178]
LLMs and IoT Software Engineering	Automating Software Development Processes	[40], [179]–[185]
	Task Planning and Automation	[8], [9], [35], [36], [183], [186]–[190]
LLMs and Sensing	Enhancing Data Collection	[5], [99], [191]–[197]
	Synthetic Data Generation	[121], [198]–[206]
	Context-Aware Sensing	[6], [7], [207]–[209]
	Sensing Application Development	[210], [211]
	Remote Sensing	[212]–[214]
LLMs and IoT Networking	IoT Network Design and Configuration	[8], [39], [51], [80], [215]
	Network Management and Orchestration	[215]–[222]
	Network Analysis	[223]–[227]
	Vulnerability Analysis and Support	[216], [228]–[233]
	Network Topology Management	[216], [234]–[237]
LLMs and IoT Data Processing	Code Generation for Data Processing	[24], [238]–[240]
	Non-Textual Data	[99]–[101], [105], [240]–[257]
	Enhancing Reasoning	[6], [258]–[263]
	Missing Data and Preprocessing	[259], [263]–[267]
	Feature Extraction	[35], [268]–[275]
	Inferences	[99], [106], [192], [260], [276]
LLMs and IoT Privacy and Security	Privacy Management	[3], [277]–[285]
	Computational Privacy	[3], [279], [286]–[291]
	Computational Security	[3], [236], [291]–[301]
LLMs and IoT Human Interaction	Graphical User Interface Agents	[302], [303]
	Enhancing Context Awareness and Adaptation	[26], [243], [304], [305]
	Personalization and User-Centric Interactions	[8], [306]–[310]
	Personal Assistants	[19], [305], [311]

However, distributing subnetworks can introduce communication overhead and impact operational trust by increasing the risk of model failures. An alternative approach is the *Mixture of Experts (MoE)* framework, where each subnetwork is treated as an expert. During inference, only a subset of

these experts is activated, enabling improved specialization and resource efficiency. MoE is particularly well-suited for agent-based frameworks, where each expert can function as a specialized LLM agent. When the experts are modeled as agents, this framework is also referred to as the Mixture of

Agents (MoA) framework [171]. The computational complexity of MoE models depends on effectively managing functional synergy among experts and selecting the appropriate experts for a given input, which can be addressed through parallelization and resource-aware scheduling for expert activation [166], [167], while task allocation techniques can optimize expert selection based on resource constraints and query characteristics [168]. In contrast, the *Lottery Ticket Hypothesis (LTH)* posits that within a large, over-parameterized model, smaller, well-initialized subnetworks referred to as “winning tickets” can be trained to achieve performance comparable to the original model [169], [312]. Identifying these winning tickets allows for the creation of lightweight subnetworks that maintain high accuracy while using fewer resources. LTH can also enhance distributed learning, having been successfully applied to optimize federated learning [170]. However, this approach can compromise operational trust; for example, LTH-based federated learning systems have been shown to be susceptible to backdoor attacks [313]. Future directions in this field include reducing communication overhead for subnetworks, developing improved mechanisms for identifying optimal subnetworks, and investigating the security and privacy threats associated with these techniques.

4) HARDWARE ACCELERATORS

Hardware accelerators are specialized components that enhance resource efficiency by enabling the simultaneous processing of multiple neural network tasks, reducing latency and power consumption [172]. The most common type of accelerator is the Field-Programmable Gate Array (FPGA), which can be reprogrammed to create LLM-specific accelerators that optimize data flow and computation for particular models. For example, custom FPGAs have been developed for LLaMA models [173]. FPGAs are especially beneficial for edge devices where multiple models may need to run concurrently. However, achieving functional synergy is difficult, as the customization process can be labor-intensive and requires specialized design solutions [174], [175]. Alternatively, Application-Specific Integrated Circuits (ASICs), also known as Neural Processing Units (NPUs), are custom chips designed to accelerate AI workloads without being tied to specific models, making them well-suited for IoT integration. They are particularly effective at reducing pre-fill delays, a major source of inference latency [176]. Recent advancements in microcontroller design also aim to enhance LLM operations on existing CPUs through optimized instruction sets and libraries. These optimizations usually focus on specific processor families, limiting their broad applicability [177]. While these techniques significantly improve LLM inference speed, the diverse architectures of IoT devices and the evolving nature of LLMs necessitate further research into functional synergy. There is also a critical need to bolster operational trust by integrating secure processors and privacy mechanisms. For example, integrating FPGAs with trusted execution environments for secure processing is highly

challenging [178]. There is a growing demand for more general accelerators capable of delivering consistent performance across various hardware platforms and LLMs.

5) COMPARATIVE EVALUATION OF DEPLOYMENT STRATEGIES

When evaluated through our analytical framework, the suitability of on-device models, collaborative distributed LLMs, and MoE architectures varies significantly across IoT scenarios. By maximizing resource efficiency, on-device SLMs (e.g., quantized sub-billion models and TinyAgent-style deployments) are well-matched to battery-powered firmware and contexts requiring high operational trust (privacy-sensitive), especially when paired with PEFT methods that avoid full-model updates. In contrast, collaborative distributed LLMs (CoLLM/Edge-LLM) leverage functional synergy via adapter layers and adaptive scheduling to achieve lower latency for complex tasks, but introduce a dependency on reliable links and modest communication overhead. MoE enhances specialization and throughput when expert activation is resource-aware. However, it increases configuration complexity and strains operational trust due to inter-agent orchestration vulnerabilities. In terms of overhead, dynamic Subnetwork selection reduces the local computational load, but introduces communication overhead that can increase total inference latency by 15-30% in low-bandwidth (BLE/LoRa) IoT networks (increases in WiFi/Ethernet networks are typically under 10%). Furthermore, Hardware Accelerators like FPGAs offer superior performance gains (1.2-1.8x) for identical kernels, but suffer from hardware heterogeneity, limiting functional synergy. A model optimized for one NPU may require significant re-engineering for another, complicating portability across diverse device fleets. Finally, operational trust risks scale with model distribution: distributing LLMs increases security vulnerabilities, especially to man-in-the-middle attacks, due to inter-node data transfers. These considerations suggest that deployment strategies should be dynamically selected based on the scenario. Lightweight quantized on-device LLMs are best suited for real-time control on duty-cycled sensors, while collaborative or MoE deployments are better for multi-step analytics and automation in LAN-connected industrial edges. Examples of IoT use cases for MoE also include autonomous vehicles and multimodal sensing.

B. IoT SOFTWARE ENGINEERING

The integration of Large Language Models (LLMs) into IoT software development is revolutionizing the creation and maintenance of applications for an expanding array of IoT devices. This section explores how LLMs can automate and enhance various stages of the software development life-cycle, including automating the software development process, and supporting task planning and automation. Note that LLMs naturally have many other uses within software engineering processes. However, these have already been

explored extensively by other surveys, such as [314], and in this section we focus specifically on IoT-specific software engineering issues.

1) AUTOMATING SOFTWARE DEVELOPMENT PROCESSES

LLMs can significantly enhance the software development process for IoT devices by generating comprehensive textual requirements and use cases based on informal information [40]. Driving resource efficiency, they can expedite code generation and fault detection, e.g., by generating dynamic trigger-action programs (TAPs) for home automation and other domains from natural language requests [179]. Other models, such as IoTcoder [180] and LLM4TAP [181], also utilize LLMs for TAP generation. In addition to code generation, LLMs can facilitate automatic validation by utilizing device-specific parameters and feedback from validation to refine the generated code [182]. LLMs can also create customized AI models or agents based on user goals and device constraints [183], [184]. Furthermore, they can be employed to bolster operational trust through defect localization and bug detection. For example, ChatDL [185] analyzes code updates and defect reports for Industrial IoT systems. By enhancing code summarization and ranking changes based on defect relevance, it can localize software errors more accurately. Future research should focus on integrating automated testing approaches to ensure robust system integration and investigating safety protocols that further enhance operational trust in IoT software development. Other directions include exploring LLMs in real-time collaborative coding environments and evaluating their potential to improve development efficiency in cross-platform development, thereby strengthening functional synergy.

2) TASK PLANNING AND AUTOMATION

LLMs excel in activity planning and workflow design, making them valuable for automating large-scale tasks across diverse IoT ecosystems [8]. Their ability to generate and execute code enables high functional synergy across devices with varying deployment characteristics, such as differing locations and environments [35], [36], [183]. Research has demonstrated IoT automation through workflow-based approaches, where LLMs act as workflow engines to ensure seamless task execution. This capability is particularly relevant in smart homes and buildings, where LLMs can orchestrate workflows that integrate diverse subsystems that lack shared spatial characteristics. LLM-powered planning can also benefit city-scale autonomous vehicle deployments or guide and enhance task coordination for robots and micro-mobility vehicles [9], [186], [187]. IoT-Together is an example of automated task planning systems [188]. It employs interactive goal management in natural language to identify user goals and generate necessary service code based on current device restrictions. Another example is DELTA, which introduces a task-decomposition method using scene graphs to develop feasible plans for robots [189]. Enhancing

LLMs' planning capabilities through task decomposition and improved reasoning methods, such as test-time inference, can further advance task planning and overall resource efficiency [190]. Despite these advantages, challenges such as latency, processing speed, and output reliability can limit the effectiveness of LLM-based automation in real-time applications, impacting operational trust [9]. Future research should focus on developing more efficient algorithms to reduce LLM latency without sacrificing output quality. Hybrid models that combine LLMs with edge computing resources may also improve processing speed and reduce overall energy consumption. Moreover, exploring ways to standardize LLM integration across various IoT devices is essential for facilitating interoperability and scalability, ensuring sustained functional synergy.

C. IoT SENSING

LLMs have powerful reasoning capabilities that can enhance the collection, processing, and utilization of sensor data. This section examines how LLMs can be used to improve and streamline sensing-related tasks in diverse IoT applications.

1) ENHANCING DATA COLLECTION

To improve resource efficiency, LLMs can enhance the data collection process by developing effective data collection plans that translate into actionable instructions for IoT devices. For example, prompts can guide the formulation of questions regarding data collection and sampling rates tailored to specific contexts, ensuring that IoT devices operate efficiently [99]. Moreover, fine-tuning LLMs can foster functional synergy by enabling the seamless integration of sensor data with environmental information or domain-specific knowledge. For instance, LightLLM [191] enhances IoT-based light sensing tasks by incorporating task-related knowledge, which is encoded through prompts to facilitate effective data fusion. Additionally, some works, such as CASIT [192], utilize LLM-based agents to preprocess IoT sensor data according to user commands, thereby reducing the volume of data transmitted. Other research focuses on leveraging LLMs for compressing sensor data in specific domains, such as smart transportation [193], employing adaptive sampling and rescaling techniques alongside Chain of Thought prompting. An et al. introduced the IoT-LLM framework, which utilizes CoT prompting to interpret complex sensor signals such as ECG and WiFi Channel State Information (CSI) by converting them into descriptive text prompts before processing. This method grounds the LLM in physical reality, significantly improving reasoning accuracy for sensing tasks compared to baseline models [196]. Additionally, multimodal LLMs (MLLMs) can be employed to dynamically control sensor power states, such as disabling power-consuming LiDAR when RGB-D cameras provide sufficient data, thereby optimizing the energy efficiency of the sensing layer [197].

LLMs also offer a potential way to bolster operational trust by establishing reputation metrics that quantify the reliability

of participating IoT devices, ensuring the crowdsourced data they provide is trustworthy and accurate [194]. Additionally, LLMs can be used to create data-driven soft sensors (or proxies) for variables that are difficult to measure directly but can be inferred from other auxiliary sensors, particularly in industrial settings [195]. LLMs can also provide knowledge about the physical world based on their embedded understanding of the environment and IoT sensor data. For example, LLMs can offer information about a user's location by providing sensor data regarding movement [5]. Future directions include optimizing LLMs for diverse sensor types to improve data accuracy and enhance context-awareness. LLM-based data compression also warrants exploration, particularly regarding the preservation of data integrity despite compression. Other potential directions include the generation of adaptive prompts based on real-time sensor feedback, and multi-sensor fusion techniques to provide deeper insights into IoT data.

2) SYNTHETIC DATA GENERATION

LLMs have the potential to significantly improve resource efficiency in synthetic data generation, alleviating issues with data scarcity and high collection costs [198]. This is particularly important in situations where collecting real-world data is impractical or expensive. Additionally, synthetic data can mitigate privacy concerns and maintain operational trust by generating information without exposing sensitive personal details. For instance, LLMs have been used to create textual descriptions of motion data [199]. Their ability to articulate human experiences can also aid in generating sensor data for human activity recognition, especially where existing datasets are limited due to variability in real-world scenarios [201]. By describing plausible ways to perform activities, LLMs enhance datasets for training machine learning models, as demonstrated in generating synthetic sensor data for human activity recognition [200] and ground pressure data for various activities [202].

LLMs can also support other domains, such as RF data generation [121]. By combining domain-specific knowledge with reinforcement learning, LLMs can effectively generate general sensor data [198]. In [203], a framework is proposed that uses LLMs to infer desired data features and select domain-specific generative models to enhance the output. GenG [204] adopts a similar strategy, fine-tuning an LLM for sensor data tasks and employing a transformer-based model for data generation. Additionally, LLMs can automate dataset creation for training small language models on devices with limited resources, such as smart homes, enabling them to plan actions based on user goals [205]. Furthermore, LLMs can optimize sample selection from existing benchmarks using adaptive sampling techniques to reduce training costs while maintaining accuracy [206]. Future directions involve refining synthetic data generation techniques to improve dataset realism and diversity, which is essential for robust machine learning. Integrating LLMs with generative adversarial networks (GANs) could enhance data quality, while evaluating

the scalability of these methods across various IoT applications is crucial for assessing their practical impact.

3) CONTEXT-AWARE SENSING

LLMs can enhance sensor data by providing additional context. For instance, they have been used to annotate raw IMU sensor data without requiring additional modalities [207], addressing scalability, cost, and privacy concerns associated with human annotators. LLMs also enrich ambient sensor data in smart homes for human activity recognition, allowing models to rely on textual data and facilitating functional synergy across spaces with varying floor plans [208]. ContextGPT [6] improves human activity recognition by leveraging LLMs to provide context based on additional raw sensor data from wearable devices and smart home sensors. This helps verify the consistency of recognized activities, thereby reinforcing operational trust. Similarly, SensorLLM [7] adds textual descriptions of variations and trends in sensor data to aid in activity annotation. Additionally, LLMs can enhance the quality of crowdsourced spatiotemporal data by incorporating high-quality textual data extracted from the web [209]. Future directions involve integrating diverse sensor inputs and ensuring that LLMs can generate real-time contextual insights to support contextual reasoning.

4) SENSING APPLICATION DEVELOPMENT

LLMs can provide a programming framework for context-sensing applications based on common sensors [210]. This framework allows natural language to serve as an interface for processing sensor data, enhancing transparency, reducing the need for programming expertise, and improving overall resource efficiency. It also facilitates the incorporation of feedback for query refinement, simplifying the application development process. Additionally, LLMs can process data collected from ambient sensors on edge devices, enabling real-time emergency detection that directly supports operational trust by providing appropriate interactions and reminders for users [211]. This capability is particularly valuable in domains such as healthcare and elderly care, where timely interventions can significantly impact outcomes. Future directions involve extending LLM programming frameworks to accommodate a broader range of sensor types and applications, and exploring ways to integrate user feedback mechanisms to create more intuitive and user-friendly context-sensing applications.

5) REMOTE SENSING

LLM agents can manage remote sensing tasks. First, they receive user commands in natural language and plan the necessary functions. Then, they execute these functions and deliver the final results to the user [212]. Additionally, LLMs can optimize the balance between sensing and communication in networks, such as uncrewed aerial vehicles (UAVs), where both functions share the same signals. By analyzing these signals, LLMs demonstrate high functional synergy

by identifying the optimal trade-off between data collection and communication efficiency [213]. This ensures that UAVs effectively gather information while maintaining reliable communication. Furthermore, leveraging multimodal large language models can enhance remote image sensing by improving visual perception and effectively integrating text and images [214]. This capability can lead to richer context and more actionable insights from remote sensing data. Future directions involve the integration of feedback mechanisms for improved operational effectiveness.

D. IoT NETWORKING

As the scale of IoT deployments increases, the complexity of the networks connecting these devices also rises. LLMs can enhance various aspects of IoT networks, including optimizing network configurations, improving resource allocation, and enhancing resilience. This section surveys the role of LLMs in enhancing IoT networks.

1) IoT NETWORK DESIGN AND CONFIGURATION

LLMs have the potential to significantly improve resource efficiency by tailoring network requirements and specifications to align with specific tasks and deployment scales [39], [215]. For instance, the design of on-device and offloading architectures can greatly benefit from LLMs. These models can select networking strategies and offloading models that optimize task distribution across the network based on device location, resource availability, and other factors [51]. This capability enhances functional synergy and communication efficiency. LLMs can also play a critical role in maintaining network-wide policies. An example is cloud execution, where policies govern data communications outside the device [80]. Ensuring operational trust is crucial here; sensitive data generated by prompts must be properly handled, either removed or obfuscated, before transmission to the cloud [8]. However, a notable drawback of this approach is that processing time and communication latency may increase. Future directions involve examining advanced algorithms and frameworks that can dynamically adapt to changing network conditions and device capabilities.

2) NETWORK MANAGEMENT AND ORCHESTRATION

Achieving functional synergy in network management is complex because IoT networks support distributed architectures that accommodate heterogeneous devices using various communication technologies. This diversity presents challenges for employing LLMs, particularly due to differences in communication protocols, data formats, and functionalities among devices. Addressing these challenges requires a consistent contextual understanding of device operations and user interactions [216]. LLMs can facilitate frameworks that guide network architecture design. This optimizes resource efficiency by offering insights into configuration trade-offs, especially in complex networks like 5G/6G and satellite systems [217]. A notable application is intent-based networking, where LLMs can autonomously generate network

configurations with minimal external intervention [218], [219]. LLMs can also be used to query and analyze data in real-time from IoT devices and network infrastructure. This supports network orchestration, dynamic task scheduling, and other critical tasks [215]. This real-time data analysis can enhance model inference and facilitate the implementation of geographically distributed LLM-based IoT network architectures [220]. Additionally, LLMs can prioritize offloading for devices that are more frequently active [221] or have higher workload demands [222]. Future directions include the integration of LLMs with intent-based networking frameworks to enhance adaptive decision-making processes. This will optimize performance while ensuring networks effectively respond to real-time changes in conditions and demands.

3) NETWORK ANALYSIS

By collecting network traffic data and translating it into natural language, custom LLMs can be developed to identify and describe different devices within IoT networks [223]. Such capabilities are essential for managing large-scale IoT systems, where accurately identifying and categorizing devices can significantly enhance resource efficiency and network responsiveness. As an example, Meyuhas et al. [224] employ an LLM to automatically detect and label IoT devices. This involves extracting and enriching textual features from network traffic, followed by applying machine learning methods for string matching. LLMs with zero-shot classification are then used to determine device functions and label them based on confidence levels. However, maintaining operational trust is a major caveat in LLM integration, as their use may introduce new vulnerabilities that affect network performance or expose configurations. For instance, while speculative decoding can enhance efficiency and accelerate LLM inference, it may lead to input-dependent variations in network packet timing and size [225]. Similarly, LLMs that analyze network data and parameters can inadvertently lead to data leakage that reveals details of the underlying network architecture [226]. Furthermore, the performance of LLMs can be impacted by factors affecting network reliability, such as interference, signal degradation, and inadequate bandwidth. This can potentially lead to incomplete or inaccurate data being transmitted to the LLM [227]. Future directions involve developing robust mechanisms to mitigate these vulnerabilities while ensuring the effective integration of LLMs in IoT networks.

4) VULNERABILITY ANALYSIS AND SUPPORT

Beyond device identification, LLMs can be leveraged to fortify operational trust through vulnerability analysis. They achieve this by diagnosing issues, describing them, and providing guidance on corrective actions to improve network resilience and performance [228], [229]. This approach has been successfully applied in optical networks, where AI agents positioned in the control layer manage the physical layer and facilitate interactions at the application layer [230]. LLMs can also optimize resource efficiency by

suggesting adjustments that enhance data transmission efficiency and increase intelligence within the network [216]. In wireless networks, in-context learning simplifies model training and hyperparameter fine-tuning [231]. LLMs have been employed for large-scale contextual analysis of network data in real-time, enabling dynamic network topology optimization [232]. In the context of 6G mobile communications, LLMs can enhance network performance and security through service personalization and comprehensive operational assessments [233]. Future directions should focus on enhancing the interpretability of LLM-based vulnerability assessments and exploring deployments that bring LLM capabilities closer to end-user IoT devices.

5) NETWORK TOPOLOGY MANAGEMENT

Many IoT deployments are highly dynamic, with devices frequently joining or leaving the network. Maintaining accurate information about the network in these environments can be challenging, even for LLMs [234]. However, by continuously examining network usage patterns, LLMs can sustain functional synergy by reducing response times to failures in network architectures, providing timely insights and practical recommendations. For example, analyzing metrics such as packet loss, retransmissions, and communication channel utilization through customized LLM-assisted analysis for specific devices and their usage contexts helps create a clearer picture of normal network behavior [235], [236]. This understanding enables targeted corrective actions, such as adjusting transmission protocols, dynamically reallocating bandwidth based on real-time usage, or optimizing routing configurations to enhance overall network performance [237]. LLMs can also bolster operational trust and network resilience by aiding in failure detection. For instance, they can establish a comprehensive baseline of expected communication patterns, which can be used to identify anomalies in heartbeat responses that may indicate potential node failures [216]. Future directions include enhancing the responsiveness of LLMs in dynamic environments, ensuring they effectively adapt to changes in network topology and device availability.

E. IoT DATA PROCESSING

The ongoing expansion of IoT is increasing the volume and complexity of data generated by interconnected devices. LLMs have emerged as powerful tools for processing this data, providing innovative solutions for code generation, data processing, and inference tasks. This section examines the diverse applications of LLMs in IoT data processing.

1) CODE GENERATION FOR DATA PROCESSING

Early investigations into the utilization of LLMs for code generation have demonstrated that GPT-3 [24], despite not being specifically trained for this purpose, can generate simple programs from Python docstrings [238]. To enhance resource efficiency, this capability allows LLMs to streamline the generation of code for modifying input datasets, creating meaningful features, preprocessing IoT sensor data, and

conducting IoT task reasoning and analytical processes [239]. By utilizing prompts that correspond to specific data-related tasks, such as data visualization, preprocessing, transformation, and analysis, LLMs can produce tailored scripts in various programming languages [240]. Each prompt typically includes a test case that describes the problem, a source file, and clear instructions for code generation. Unlike natural language samples, which are typically evaluated by human evaluators or heuristic algorithms, the generated scripts undergo validation to ensure they run successfully and meet specified requirements. Popular benchmarking frameworks, such as HumanEval [238], measure the functional correctness of these scripts based on their success rate on the first attempt. Future directions include enhancing the robustness of LLM-generated code to bolster operational trust, integrating feedback mechanisms for iterative improvements, and extending code generation capabilities for more complex IoT applications, such as real-time data processing and machine learning model deployment.

2) NON-TEXTUAL DATA

LLMs were initially designed for processing and generating text, while IoT systems incorporate diverse sensor modalities, including time series, spatial data, voice, and images. Achieving functional synergy across these diverse data types is a primary focus of recent advancements. These developments have enabled LLMs to better handle various IoT sensor data types, such as spatio-temporal data [241], [242], raw IMU data [99], [101], [243], sensor data from smart homes [105], and time-series data [244], [245], [246], [247], [248], [249]. Since LLMs can infer from sparse and incomplete data, several methods have been proposed to effectively represent sensor data. One approach is to include raw sensor data within the prompt [250], [251]. Another approach involves transforming spatio-temporal and time-series data into textual representations, such as strings of numerical digits or sequences of tokens. This frames time-series forecasting as a next-token prediction task [252], [253]. Additionally, using statistical summaries instead of full raw data can reduce input size and improve response efficiency by minimizing the need for LLMs to process large datasets [100]. More recently, TimeSeriesScientist [257], an LLM-driven agentic framework, leverages four agents for augmentation and preprocessing, self-planning, forecasting and validation, and reporting to provide an interpretable and transparent time-series forecasting process. With the evolution of Large Multimodal Models (LMMs) like GPT-4V(ision) [254], LLMs have also been extended to analyze images through multimodal plugins such as Bing Image Search [255]. This advancement enhances the understanding and inference of the semantics of audio and visual signals. LLMs can also assist in selecting the best analysis tool for data based on user conversations [256]. Marble [65] utilized multimodal inputs, such as images, videos, and text to generate 3D worlds and provide spatial intelligence functionalities. However, accommodating errors encountered in operations on mixed data types, such

as performing arithmetic operations between numbers and strings, remains a challenge [240]. Future directions include improving sensor data representation to enhance fusion, modeling, and reasoning, as well as developing more robust methods for representing and processing mixed data types.

3) ENHANCING REASONING

To improve the reasoning capabilities of LLMs, additional context regarding time-series inputs and task instructions can be provided in natural language [258]. Large language models can use advanced natural language processing techniques to understand and analyze complex patterns that change over time and space. By first training them on non-textual data and then fine-tuning them for specific tasks, LLMs can improve their accuracy in predicting future values in time-series forecasting [258], [259]. More advanced techniques involve integrating domain knowledge into neuro-symbolic models, which can significantly improve activity recognition accuracy [6]. In addition to these strategies, LLMs can achieve high functional synergy by constructing complex multimodal representations that fuse data streams from various sensors [260]. Furthermore, incorporating encoders designed specifically for time-series or spatio-temporal data has shown significant promise. For instance, Li et al. [261] utilized neural networks to effectively encode temporal dependencies in spatio-temporal data. Moreover, training or fine-tuning LLMs specifically for time-series data proves to be an effective strategy, as demonstrated by Bian et al. [262], who first trained an LLM on patch-based predictions to learn general time-series features across domains before fine-tuning it for multi-patch forecasting. In addition to text-based representations, LLMs can process structured data, such as graphs or tables, through specialized embedding techniques that capture relational and hierarchical information [263]. Future directions include exploring more sophisticated multimodal models that integrate and support diverse data types.

4) MISSING DATA AND PREPROCESSING

Missing data is a common and challenging issue in IoT data analysis tasks, and LLMs are increasingly employed to evaluate and impute these missing values [263], [264]. Addressing these gaps is essential for maintaining operational trust in the generated insights. The approach leverages the pre-trained knowledge of LLMs to understand complex patterns and relationships within the data, enabling the generation of accurate imputations for missing values. For instance, LLMs integrated with a graph attention module can effectively capture spatial and temporal dependencies, resulting in more precise imputations for spatiotemporal IoT sensor data [263]. This capability not only improves data quality but also enhances the reliability and operational trust of subsequent analyses and predictions. Additionally, LLMs can assist in preprocessing time-series data by employing techniques such as padding and tokenization to manage variations in sequence length and create meaningful sequences [265]. To facilitate the integration of information from multiple time scales and

improve prediction accuracy, specialized frameworks like LLM4TS have been developed [259]. Furthermore, task-aware reduction methods that focus on reducing input data for LLMs based on downstream tasks [266] are used to enhance LLM performance by prioritizing relevant data. The Flash-Fusion [267] system uses a similar approach to retrieve query-relevant sensor data by using a planner with a semantic understanding of collected data and the user query. Future directions involve extending these frameworks toward advanced imputation techniques capable of handling missing values while also detecting and processing outliers and anomalous data points.

5) FEATURE EXTRACTION

Effective data analysis relies on features that accurately characterize the target problem and correlate with desired outputs. LLMs can leverage their pattern learning capabilities to identify meaningful correlations, automating feature generation to boost resource efficiency and facilitating the explainability of features to ensure operational trust [268], [274]. Effective feature extraction requires well-structured prompts that guide the model in extracting relevant features. These prompts typically include task descriptions, expected inputs or outputs, and relevant examples [272]. They can be developed from initial candidates suggested by LLMs [269], guided by chain-of-thought reasoning [270], or constructed based on insights from other prompts [271]. Domain-specific prompts can further refine LLM outputs to generate structured representations, extracting meaningful insights for improved forecasting [275]. For instance, prompts informed by smart home routines can guide LLMs to propose features tailored to user behaviors [35]. Effective feature extraction can also be achieved through few-shot and zero-shot prompting by utilizing the embedded knowledge of pre-trained LLMs [272]. Additionally, LLMs facilitate interactive feature exploration via natural language queries, allowing users to engage intuitively with the data and extract insights without specialized technical knowledge [273]. With continuous learning capabilities, LLMs can adapt to new data and user feedback, enabling the extraction of more relevant and interpretable features. Future directions include exploring the potential for LLMs to collaborate with domain experts in real-time, refining feature extraction processes to ensure that derived features meet practical application needs.

6) INFERENCES

LLMs are increasingly utilized for IoT inference, particularly in causal analysis of sensor data to understand how various parameters influence specific outcomes and to improve decision-making [106]. For example, by supplying LLMs with detailed information about data collection methods, categories of human actions, and well-crafted prompts, they can analyze IMU sensor data to recognize common activities such as sleeping, walking, standing, or bicycling [99], [260]. This capability provides a deeper understanding of human behavior and its impact on outcomes. When models

for unseen tasks are unavailable, GANs can be combined with LLMs to make inferences without the need to train new models from scratch [276]. Additionally, LLMs enhance data analysis and inference capabilities through natural language commands, allowing users to interact with the system using everyday language for nuanced data exploration. The effectiveness of inference can be further improved through collaborative LLM agents [192]. Seamless communication and cooperation among these agents maximize functional synergy, enhancing the understanding of human actions and optimizing the operational efficiency of IoT systems. Future directions include investigating the integration of LLMs with real-time feedback loops from IoT devices to continuously refine inference models.

F. IoT PRIVACY AND SECURITY

IoT ecosystems pose significant security and privacy challenges that LLMs can help address. This section explores the potential LLMs can bring to improve security measures and enhance privacy protection. Security and privacy risks in LLM integration were covered in Section IV.

1) PRIVACY MANAGEMENT

The natural language capabilities of LLMs support the integration of privacy policies, user preferences, and regulatory requirements [280]. For example, users can utilize contextualized prompts to customize their privacy settings and understand the implications of granting permissions. By providing transparent natural language interfaces for users to control their interactions based on personal preferences, LLMs directly bolster operational trust [277], [278]. These tools allow users to specify which sensor data they are willing to share, helping them make informed decisions about their data [279]. These tools can also assist in assessing regulatory compliance and extending legal protections, such as those offered by GDPR (General Data Protection Regulation) [280], [285]. LLMs also facilitate the implementation of privacy-enhancing technologies (PETs), which can be integrated as dedicated AI agents for tasks like anonymization and data minimization [281], [282], [283], [284]. Finally, LLMs can generate tailored educational content to inform users about best practices for data protection, such as changing default passwords or enabling multi-factor authentication [3]. Future directions include the development of user-friendly interfaces and educational tools that empower users to manage their privacy more effectively.

2) COMPUTATIONAL PRIVACY

LLMs can also facilitate the integration of privacy protection mechanisms into IoT environments. For instance, LLMs can help evaluate k-anonymity to reduce vulnerabilities to re-identification attacks [286], [287]. They can support differential privacy by determining appropriate perturbation mechanisms or the level of noise that should be added [288], [289], [290]. Similarly, LLMs can automate the de-identification process by analyzing collected data and

applying techniques for anonymization. They can generate unique tokens for pseudonymization and suggest methods for generalization or data swapping [287], [291]. Given the vast amounts of data in IoT deployments, identifying effective protection mechanisms is crucial. While LLMs enhance privacy integration and operational trust, it is important to recognize that training these models often involves sensitive data related to users' behaviors and environments [3], [279]. Thus, the benefits of LLMs must be carefully balanced to ensure resource efficiency is not compromised by the risks of sensitive data leakage. Future directions involve investigating methods to minimize the exposure of sensitive data during LLM training.

3) COMPUTATIONAL SECURITY

Similar to privacy, LLMs can play a key role in improving security by explaining security principles, providing guidance on enhancement measures, and offering insights into potential threats [3], [292]. For example, guided prompts can facilitate integrity checks on collected data for anomaly detection considering the type of data and historical records [293]. By analyzing data usage patterns, LLMs can help enforce security policies for data access and sharing, and assist in firewall configuration management by advising on application-specific traffic management requirements. LLMs can recommend suitable security mechanisms, including symmetric or asymmetric encryption standards, cryptographic hashing methods, and multi-factor authentication strategies tailored to IoT contexts [291], [294]. LLMs have been used to identify vulnerabilities, such as risks of denial-of-service (DoS) attacks, by analyzing usage patterns to detect abnormal traffic [295]. Their capabilities demonstrate high functional synergy by monitoring security threats across various IoT layers. For example, LLMs can reduce the risks of backdoor attacks by scanning for vulnerabilities and ensuring firmware integrity checks [296], [297]. Another example is traffic analysis to secure data packets during transmission, generating configuration scripts for security protocols like WPA, IPSec, TLS, or VPN [236]. On the other hand, LLMs are vulnerable to modality attacks via prompt-based injection, especially when using multimodal sensor data. To reduce this vulnerability, frameworks such as JailGuard [300] and E^2AT [301] can be used to evaluate prompts or enhance the training process. Future directions include exploring the integration of LLMs with real-time threat intelligence solutions. Additionally, research should focus on improving the operational trust of federated and distributed learning by identifying potential vulnerabilities in data sharing [315]. Finally, evaluating vulnerabilities associated with device usage and context, including communication and computation overheads, remains crucial for optimizing resource efficiency [298], [299].

Practical mitigation of ethical and privacy concerns requires a layered approach. At the data level, various data minimization methods can limit the exposure of

sensitive attributes. These include prompt and log redaction, short-context policies, and selective sensor fusion. Privacy-enhancing technologies (PETs) can be integrated into data processing pipelines to control identity leakage. For example, PET methods such as k-anonymity for spatiotemporal traces [286], calibrated differential privacy for time-series analyses [289], and pseudonymization with token vaults [291] can be utilized. At the execution level, trusted environments, encryption of sensitive layers within Trusted Execution Environments (TEEs) [316], and encrypted KV caches [317] reduce the risk of on-device compromise. Furthermore, mutual authentication hardens multi-agent or distributed inference. Hallucination control methods such as contrast decoding [135], localized fine-tuning [124], LLM-as-a-judge audits [133], and cross-check consistency among multiple models [140] strengthen operational trust. Finally, governance artifacts (such as model cards and risk registers) and user agency modes (manual, automated, and hybrid interactions) promote transparency and safeguard autonomy [310], improving alignment with emerging regulatory frameworks.

G. HUMAN INTERACTION IN IoT

LLM capabilities offer numerous opportunities to enhance human interactions in IoT environments. This section explores these interactions, focusing on their contributions to user interface design, context awareness, personalization, and intelligent personal assistants.

1) GRAPHICAL USER INTERFACE AGENTS

LLMs can significantly enhance the design of graphical user interfaces (GUIs) by developing prescriptive models of user interactions and descriptive models of human behaviors. This approach improves resource efficiency by reducing the reliance on extensive, costly physical user studies [302]. Traditional IoT design often relies on user studies that capture behaviors limited to specific populations, which may not account for diverse or exceptional contexts. For instance, findings from a suburban study may not apply to urban users who leave smart lights on for security. This highlights the need for more inclusive design approaches that consider a wider range of user behaviors. By integrating LLMs with human behavior descriptors, designers can simulate comprehensive human-machine interaction scenarios, including uncommon behaviors, thereby maximizing functional synergy across diverse user bases [303]. Prompt templates can be created to gather contextual human behavior inputs, such as, “Given this [scenario X], what is [behavior Y] of [user Z]?” in home energy management systems [302]. Future directions involve assessing the effectiveness of these templates across diverse demographics and contexts to refine models and improve real-world applicability.

2) ENHANCING CONTEXT AWARENESS AND ADAPTATION

IoT systems function in dynamic environments where user needs can change rapidly. To adapt effectively and deliver

relevant services, LLMs leverage functional synergy by continuously learning from user interactions to improve system responses. For instance, an LLM can analyze data from a smart thermostat to optimize heating and cooling based on user preferences. Multimodal LLMs like LIMU-BERT [243] and Llama [26] can interpret sensor data. They can then generate human-like responses to user inquiries about activities tracked by wearables [304] or assess sleep quality using data from smartwatches [305]. A key area for future research is maintaining operational trust when LLMs must resolve conflicting data from multiple sensors to ensure system accuracy. This involves developing algorithms that prioritize sensor data based on context and user preferences. Simultaneously, these algorithms must address the implications for user privacy, usability, and system reliability.

3) PERSONALIZATION AND USER-CENTRIC INTERACTIONS

LLMs can foster high functional synergy by providing a unified interface that interprets user intents across various devices using natural language processing and voice commands [306]. This simplifies communication, enabling users to control devices seamlessly. Users can say, “Turn off all lights and set the thermostat to 24 degrees,” rather than navigating multiple apps. Other studies, such as Ideal-LLM [318], also enhance multilingual adaptation of LLMs for speech-to-text tasks. End-to-End speech LLMs make interactions more user-friendly. They perform automatic speech recognition for many languages and dialects while also generating audio responses [319], [320], [321]. Furthermore, LLMs can personalize interactions by learning individual preferences and behaviors over time [8]. This potentially helps to make IoT systems more sophisticated and encourages wider adoption. To improve personalization, LLMs can generate summaries tailored to user preferences, employing reward functions to refine summarization strategies [307]. A multisensory multi-task adapter layer can also enhance the integration of diverse data types and tasks, leading to more efficient interactions in IoT systems [308]. However, the autonomy of LLMs in these interactions raises concerns about user control over data and decision-making, which can diminish operational trust and user agency [309]. To mitigate these concerns, researchers are developing automated, manual, and hybrid interaction modes. These allow users to choose the approach that best suits their needs. Users can also initiate specific LLM-controlled sequences on request [310]. Future research should focus on user-centric frameworks to assess the effectiveness of LLM-based interaction modes on user satisfaction and to explore the advantages and disadvantages of LLM-driven personalization.

4) PERSONAL ASSISTANTS

The diversity of device ecosystems and differing user interfaces can create confusion and lead to underutilization as users often struggle with complex commands. LLMs can

TABLE 4. Challenges of integrating LLM technologies into IoT layers.

Challenges	Devices	Software Engineering	Sensing	Networking	Data Processing	Privacy & Security	Human Interaction
#1. IoT-Specific LLMs	Processing architectures for non-textual data	Development of specialized LLM frameworks	Understanding spatial and time-series data	Network variations in data transfer	New algorithms for diverse data types	Concerns with sensor data integration	User-friendly interfaces for customization
#2. Penetrative AI	Advanced HW for continuous sensing	Integration of LLMs with existing software	Improved sensing technologies	Robust networking for real-time data	Data processing for massive data	Real-time data privacy and security concerns	Semantic navigation interfaces
#3. High-Level Reasoning & Cognitive IoT	Adaptation for collaborative reasoning	Support for interoperability	Enhanced internal/external state sensing	Connectivity for collaborative reasoning	Real-time adaptability in data analysis	Security in collaborative environments	User interfaces for complex interactions
#4. LLM-Enabled Semantic Comms	Hardware support for LLMs	Model training and adaptation	Sensing for accurate data interpretation	Optimization for reduced data transmission	Ensuring semantic interpretation	Security in semantic communications	Interfaces for agent-based communication
#5. Quantum-Enhanced LLMs	Quantum-IoT Hardware Integration	Software for quantum-enhanced models	Support to quantum sensors	Networking for quantum data transfer	Complexity in managing large data volumes	Security of quantum-enhanced models	Quantum-Enhanced UIs
#6. Human-Robot Interactions	Hardware for advanced robotics	LLMs for understanding natural language	Sensing for contextual awareness	Networking for real-time communication	Analysis of user interactions	Ethics and safety in robotic operations	Action languages for intuitive interactions
#7. Secure Execution and LLMs	TEEs for secure execution of LLMs	LLMs adaptation for resource constraints	Multi-device sensing implementation	Network partitioning for model execution	Efficient models for constrained environments	Protecting sensitive data in LLMs	Efficient interaction with secure models
#8. Hyperautomation	Hardware for supporting hyperautomation	LLM integration in automation software	Sensor LLM sampling coordination	Networking for seamless automation	Contract validation and compliance	Safeguarding against legal risks	Natural language interfaces for contract creation

be integrated with personal data, devices, and services to streamline tasks, improving resource efficiency by reducing redundant device interactions and by planning multi-device actions in a single workflow (e.g., “lock doors, turn off lights, set thermostat”), lowering the total number of requests and reducing network traffic. Additionally, LLMs can aggregate data from devices and offer insights into their operation. LLMs can be used to schedule tasks based on contextual factors to optimize resource usage through dynamic resource allocation and energy efficiency. Personal LLM agents can extend Intelligent Personal Assistants by leveraging advanced semantic understanding and reasoning capabilities to enhance user experiences, automate repetitive tasks, and provide personalized recommendations [19]. Similarly, LLM-powered personal health assistants exhibit functional synergy by connecting with wearable health monitoring devices to deliver tailored consultation services [305]. This bridges the gap between users and healthcare professionals, including virtual and physical doctors [311]. Future research should focus on evaluating how effective different LLM architectures are when used for personal assistance. It also includes exploring user perceptions and acceptance of these technologies across multiple IoT application domains, including healthcare, home automation, and personal productivity.

VI. FUTURE DIRECTIONS AND GAP ANALYSIS

Our survey has provided a comprehensive overview of the various ways in which LLM integration is utilized within

modern IoT ecosystems. As the research landscape evolves, new challenges emerge, and some existing ones remain unaddressed. In this section, we discuss emerging directions and trends, as well as the open gaps identified by our survey. The trends and their associated challenges across different IoT layers are summarized in Table 4, while the gaps in current research are outlined in Table 5.

A. FUTURE DIRECTIONS AND EMERGING TRENDS

We next outline some high-potential directions for future research that warrant further exploration and will benefit the broader IoT ecosystem. These are not intended as exhaustive rather as a sample of diverse research areas where further research is needed to improve the integration of LLMs and IoT. For convenience, these topics are also summarized in Table 4.

#1 IoT-Specific LLMs. While current LLMs excel at processing linguistic and visual data, their potential in handling diverse data modalities within IoT ecosystems remains largely untapped. IoT devices generate varied data types, notably spatial and time-series data, which require specialized processing techniques for accurate analysis. For example, spatial data requires an understanding of geometries, whereas time-series data relies on recognizing temporal relationships. Although initial efforts have been made to develop specialized LLM frameworks as also discussed during our survey, for tasks like geographic reasoning [322], [323] and wearable sensor data analysis [61], the limitations

TABLE 5. Mapping of research gaps to IoT layers.

Challenges (Research Gaps)	Device	Software Engineering	Sensing	Networking	Data Processing	Privacy & Security	Human Interaction
Data Modalities and Robustness	✓		✓		✓		
Real-Time Context Awareness and Dynamic Adaptation		✓	✓		✓		
Closed-Loop Learning and User Feedback		✓			✓		✓
Privacy, Security, and Ethics	✓	✓	✓			✓	✓
Interoperability, Scalability, and Cross-Domain Adoption	✓	✓		✓			✓

of LLMs in processing non-textual and non-visual data types are still not well understood. In real-world scenarios, LLMs must address the inherent unreliability of IoT data, such as missing values or anomalies in sensor data. While traditional imputation techniques exist, the reasoning capabilities of LLMs offer a novel layer for inferring missing context or identifying the underlying mechanisms of data loss, thereby selecting the most suitable imputation technique for a given situation. Furthermore, existing acceleration techniques may not be optimal for these specialized architectures, necessitating new benchmarks and evaluations specifically for real-world IoT data modalities [324]. Further research is essential to identify these limitations and to create algorithms and models tailored specifically to these data types. There is also a need for IoT-specific foundation models that support various data modalities. This requires curating datasets that accurately reflect the unique characteristics of IoT environments, addressing storage, processing, security, and privacy needs.

#2 Penetrative AI. Penetrative AI extends LLM capabilities to interpret and articulate real-world dynamics through continuous environmental sensing. By creating digital representations akin to digital twins, opportunities arise for innovative semantic applications. For example, semantic localization can identify specific IoT deployments [325], while semantic navigation can utilize IoT maps for enhanced wayfinding [326]. Combining these capabilities with LLMs facilitates intuitive interactions, e.g., through the dynamic generation of semantic navigation maps using natural language. At the same time, achieving this vision requires advances across all IoT layers. While current sensing technologies effectively characterize people and objects, further advances are needed to detect changes in internal and external states. By constructing behavioral profiles and aligning them with the emotional state characteristics of users [327], we can enhance interactions between humans, digital environments, and the physical world, fostering a more seamless and intuitive integration [328]. However, achieving high-fidelity representations requires integrating diverse sensors, which may be distributed and generate massive amounts of data. Efficiently processing this data is a considerable challenge, as is coping with network variations that can affect data

transfers. Additionally, the use of real-time data raises significant privacy and security concerns.

#3 High-Level Reasoning and Cognitive IoT. IoT ecosystems are characterized by a high degree of heterogeneity in operating environments, devices, users, and software. LLMs enhance the integration of these devices by improving interoperability, which previously required significant effort to adapt solutions from one domain to another [329]. Furthermore, the advancing reasoning capabilities of LLMs enable the generation of high-level contextual descriptions that individual data sources alone cannot provide [241], [260]. This capability allows for a more nuanced understanding of context, reducing the reliance on tailored domain-specific rules or descriptions that were essential prior to the advent of LLMs [330]. While these developments showcase LLMs' potential to enhance comprehension, these works have also highlighted significant variations in the reasoning support offered by different models [260]. However, significant research gaps remain when transitioning these reasoning capabilities to real-world deployments. A major challenge is the reliability-latency trade-off. High-level reasoning for complex tasks creates significant computational overhead. This makes it difficult to achieve the millisecond-level response times needed for safety-critical systems [331]. Recent studies have identified execution drift as a major real-world hurdle. In non-deterministic environments, even a minor sensor error can cause a cascade of logical errors in the model's reasoning chain, leading to physically implausible actions [196]. Furthermore, the non-deterministic nature of LLM outputs poses a risk in heterogeneous environments where strict logical consistency is required to manage diverse IoT actuators safely [20]. These discrepancies call for further research to explore how LLM-based reasoning can be better integrated into IoT applications, particularly in enabling real-time adaptability to changing conditions and user needs. Such adaptability is vital for ensuring that IoT systems can react to dynamic environments. Another promising avenue is the establishment of reasoning capabilities across a network of IoT devices. This interconnectedness could facilitate the emergence of a Cognitive IoT, where devices collaboratively reason and analyze the data they collect. By fostering collaboration among IoT devices, these systems can engage in complex decision-making processes leveraging shared knowledge and insights. Such advancements would significantly enhance the functionality and intelligence of IoT applications, enabling them to operate more autonomously and effectively.

#4 LLM-Enabled Semantic Communications. Future IoT communication systems are expected to prioritize semantic communication where the focus shifts from raw data transmission to meaning and context. This approach reduces the need to send large data streams, alleviating demands on the network. LLM-enabled communications extend this by leveraging LLMs to support data interpretation and bridge semantics across communication endpoints [332]. Semantic communication is expected to be particularly useful

for agent-based approaches where it can enhance both human-agent and agent-agent communication [333]. While semantic communication can leverage the contextual understanding and reasoning capabilities of LLMs, the limitations of the underlying knowledge model also constrain what such communications can offer [334]. There have been some efforts to enable LLM-enabled semantic communication, such as LLM-Twin [334] which targets semantic communication for digital twins. However, ensuring the accuracy and reliability of semantic interpretation across diverse devices and domains requires robust model training and continual adaptation. Similarly, to leverage LLMs, we need to deploy them on all devices, which depends on advances in model compression and edge computing [334]. Despite these challenges, the potential of semantic communications is vast and they can enable a new generation of more intelligent and context-aware networks.

#5 Quantum-Enhanced LLMs. Quantum computing offers a promising pathway to achieving superior performance and facilitating large-scale learning of complex machine learning models, including LLMs [335]. Although current quantum circuits are not yet sufficiently mature for widespread, mainstream application, there are already opportunities to leverage existing quantum circuits to enhance LLM performance. For example, the weight matrices in the deep layers of LLMs can be replaced with variational quantum circuits, potentially reducing the memory footprint of LLMs without compromising accuracy [336]. Despite the exciting possibilities that quantum-enhanced LLMs present, existing research has primarily focused on improving or accelerating computations for current LLM architectures, rather than exploring the integration of quantum circuits into IoT ecosystems. This gap highlights an opportunity for further investigation into how quantum computing can support the deployment and integration of LLMs within IoT frameworks. The potential applications of combining LLMs and quantum computing in IoT ecosystems are vast, particularly in scenarios involving the processing of large volumes of data or managing complex models characterized by a high number of parameters.

#6 Human-Robot Interactions. Robotics platforms are increasingly becoming integral components of our daily lives, serving as specialized examples of IoT systems. They are already being utilized in a variety of real-world applications, including delivery robots, manufacturing automation, healthcare assistance, and domestic support. Moreover, the emergence of advanced humanoid robots capable of mimicking human gestures, behaviors, and expressions—signals a significant leap in capabilities. LLMs are seen as a potential enabler for next-generation robotics platforms [337]. These models empower robots to process and understand natural language, enabling users to communicate their needs and preferences seamlessly. LLM integration can also enhance a robot's ability to interpret contextual cues and emotional subtleties. This leads to interactions that can be more

empathetic and highly responsive. As robots become adept at grasping nuanced language, they can adjust their behaviors in real-time, providing personalized assistance. As the field continues to evolve, it is crucial to address the challenges associated with LLM integration to ensure that robotic systems are safe, ethical, and beneficial to society. Key considerations include the development of techniques for hallucination detection and critical threat recognition, which are essential for maintaining safe robotic operations. There is also a need to create more descriptive action languages that provide clear behavioral models for robots to follow. Early research indicates that such descriptions can significantly enhance the responsiveness of robotic platforms, enabling complex tasks like the autonomous manipulation of objects [338]. Furthermore, the successful integration of LLMs into robotics necessitates ongoing advancements in system design and LLM execution to guarantee that interactions with LLMs are fast and resource-efficient.

#7 Secure Execution and LLMs. Commercial LLMs typically operate within data centers or other infrastructure that is physically secure. This contrasts sharply with IoT devices that can be deployed in unsecured environments, making them vulnerable to physical access by malicious actors and raising concerns about the security of both the data and the AI models on these devices [339]. To address these challenges, IoT devices increasingly integrate Trusted Execution Environments (TEEs), which provide tamper-resistant execution through secure cryptoprocessors and encryption. However, adapting LLMs to run within TEEs is complex due to their limited memory and processing capabilities, which can hinder the efficient execution of large models [340]. A promising approach involves identifying sensitive components of the model that require heightened protection, allowing these parts to run in the TEE while leveraging external GPU and CPU resources for less sensitive components [316]. Nonetheless, this may lead to potential leakage of sensitive information, such as key-value (KV) cache values, unless adequately protected [317]. Another solution is to partition the model across a network, enabling each device to execute only the portions that fit within their TEEs. Advances in LLM efficiency are crucial for adapting LLMs for TEE execution as they facilitate model size reduction and improve execution efficiency in resource-constrained environments. Continued research is essential to enhance the safety of LLMs and the data they process in IoT-rich environments, ensuring secure deployment.

#8 Hyperautomation. LLM integration into IoT ecosystems can support scaling up automation toward *hyperautomation* where business processes and workflows are heavily automatized [341]. While there are currently technologies for automating business processes, such as the use of smart contracts that operate on top of Blockchain or another distributed ledger technology [342], these remain somewhat limited due to the complexity in establishing the contracts and verifying them. LLM technology can facilitate and automate contract

creation, offering natural language interfaces for specifying the conditions that the contracts should satisfy, and can help translate the requirements into executable code. This streamlines the creation process, and decreases the level of expertise that is needed to create them. The advanced reasoning capabilities of LLMs are also important for large-scale automation as they facilitate validating contracts and support conflict resolution. The validation capabilities can also be extended to ensure continued compliance with legal and regulatory frameworks, safeguarding industries against legal and compliance risks and minimizing the need to recreate contracts as regulatory frameworks change. While the benefits of LLMs to scaling up workflow automation are clear, further research is needed to better integrate LLMs with current automation processes. Among others, this involves the creation of natural-language interfaces for supporting contract creation, the establishment of adversarial models that can be used to analyze pitfalls of LLM-based contract validation, as well as improved LLM reasoning capabilities that can support contract validation. There is also a need to study how LLMs and automation processes can be further enhanced in specific fields, such as smart grid management or logistics.

B. GAP ANALYSIS

LLM integration presents a transformative opportunity to enhance every aspect of IoT ecosystems. Yet, as we delve deeper into the current landscape, we can identify several challenges that remain unaddressed. Below, we discuss these research gaps in detail, summarizing their implications and the need for further investigation, as also illustrated in Table 5.

1) DATA MODALITIES AND ROBUSTNESS

Current software and hardware optimization techniques primarily focus on enhancing the efficiency of existing LLM architectures for IoT devices. However, a gap remains in native support for complex time-series and spatial data modalities. There is a lack of standardized benchmarks and evaluation frameworks tailored for multimodal IoT inputs. Furthermore, while LLMs offer novel ways to handle data unreliability and to address missing values or anomalies in sensor data [324], the field currently lacks robust methodologies to systematically validate LLM-driven imputation and anomaly detection across diverse sensor types.

2) REAL-TIME CONTEXT AWARENESS AND DYNAMIC ADAPTATION

The ability to analyze multi-sensor inputs and generate real-time insights about the operating environment is crucial for enabling LLMs to support context-awareness. Achieving this capability demands a tight integration of LLMs into applications designed to process and analyze sensor data. Currently, there is limited architectural support for such integration. Real-time feedback loops that enable the continuous adaptation of LLM parameters based on immediate data are critical for ensuring accurate responses in dynamic IoT environments. These feedback loops are particularly vital in

safety-critical scenarios. For instance, IoT tools designed to assist emergency responders must provide instant access to relevant information, while factory workers depend on precise guidance to maintain safety and efficiency. Implementing effective feedback loops requires algorithms capable of rapid data stream analysis to transform data into actionable inputs for the dynamic adjustment of LLM parameters. Furthermore, there is a critical need to support model adaptation in environments that change dynamically, such as detecting data drift and deploying agents that focus on specific sensors or sensor combinations as the available data evolves.

3) CLOSED-LOOP LEARNING AND USER FEEDBACK

A significant challenge remains in bridging LLMs with complex IoT data-analysis pipelines to facilitate continuous learning. While feedback mechanisms are well-established for textual data, IoT applications often operate in contexts where traditional user input is difficult or even dangerous. Fields such as recommender systems and virtual reality have explored implicit feedback methods [343], [344], such as capturing and analyzing user gaze patterns or monitoring interaction durations. Similar mechanisms are needed for LLMs integrated into IoT applications operating in complex real-world settings, such as industrial manufacturing, healthcare, mining, or military applications. Further research is needed to develop user-friendly methods for gathering implicit feedback and to seamlessly integrate these mechanisms into the LLM life-cycle. By doing so, we can refine and evolve model outputs to enhance both the effectiveness and user experience of LLM-powered IoT solutions.

4) PRIVACY, SECURITY, AND ETHICS

The heterogeneity of data and devices in IoT environments makes ensuring privacy and security highly challenging. LLMs trained on diverse IoT data could support auditing these systems by analyzing real-time data streams to detect outliers, such as targeted poisoning attacks, and generating security reports. However, the integration of LLMs also introduces severe privacy risks, such as data leakage and prompt injection [43]. A critical ethical concern is that innocuous sensor data can inadvertently expose sensitive information when analyzed by LLMs. This is particularly concerning when seemingly benign data are combined with other data to reveal sensitive personal or confidential information such as socioeconomic status or confidential health details. To maximize the benefits while mitigating these risks, it is essential to provide strong governance frameworks. LLMs must be capable of performing meaningful analyses on differentially private data or k-anonymized datasets, ensuring that individual privacy is preserved while still deriving valuable insights. Furthermore, additional research is needed to prevent the leakage of sensitive information in LLM outputs and to integrate privacy mechanisms directly into the outputs generated by LLMs. Similarly, there is a need to integrate with privacy management interfaces that can enable configuring which data types

to consider and which to exclude. Finally, as existing ethical guidelines often fail to cover the complex chaining of AI models [345], future research must focus on dataset curation, regular audits, and privacy management interfaces that allow for granular control over data inputs. Developing these robust guidelines is vital to ensure that the integration of IoT data with LLMs remains both ethical and secure.

5) INTEROPERABILITY, SCALABILITY, AND CROSS-DOMAIN ADOPTION

While the diverse frameworks for LLM integration into IoT ecosystems are promising, significant gaps remain in deploying them in practical implementations. IoT ecosystems often rely on legacy systems that cannot handle advanced AI/ML techniques. Furthermore, even modern hardware may be unsuitable for LLMs unless specifically designed for that purpose. Different subsystems of a larger IoT deployment can also be heavily fragmented and leverage different transmission protocols, security mechanisms, and other components. To address these hardware constraints, there is a growing shift toward deploying Smaller Language Models (SLMs) or specialized, domain-specific agents. However, this transition introduces new complexities in scalability and coherence. As multiple models and agents are chained together or operate in tandem across different domains, ensuring their combined responses remain interoperable and free from hallucinations is critical. To harness the full potential of these systems, future research must establish robust protocols and user-friendly frameworks. These should facilitate cross-domain interactions, including legacy solutions. Additionally, they must allow LLMs to scale by customizing parameters to meet the specific constraints of host devices.

6) CONCLUDING REMARKS

For real-world deployment, there is a need for standardized, scenario-driven benchmarks that couple accuracy with system metrics (latency/energy/throughput), privacy leakage (under PETs and TEEs), and robustness (to sensor noise, drift, and adversarial inputs). Moreover, cross-layer orchestration remains underexplored. Agents should reason over device constraints, security policies, and network state, not just task semantics. Tooling is lacking for portable accelerators and encrypted KV-caches at scale. Finally, human interaction requires stronger attention to transparency and user agency, including manual/automated/hybrid modes and audit trails that can be reviewed by operators and end users.

VII. CONCLUSION

The integration of LLMs with IoT represents a transformative shift in how we manage, interpret, and interact with vast networks of interconnected IoT devices. This survey has systematically explored the roles that LLMs can play across the IoT architecture, evaluating the convergence of these technologies through the lenses of resource efficiency, functional synergy, and operational trust. As the number of connected devices grows, the complexity of data management

and contextual interactions becomes more pronounced. Traditional methods often fall short, necessitating the advanced, LLM-driven frameworks, as critically surveyed in this work.

Our comprehensive review of over 300 articles demonstrates how LLMs can enhance IoT functionalities. These models improve interpretation, enable natural language interactions, and facilitate automated, data-driven decision-making. However, successful integration requires navigating critical challenges. We have highlighted the need to balance resource constraints against model complexity and the importance of maintaining operational trust through robust privacy and security mechanisms. Our findings show that while the field has made significant strides in functional synergy, transitioning these capabilities to real-world deployments remains challenging. We identified key research gaps, including the need for specialized architectures for diverse IoT data types and the emerging potential of technologies like quantum computing to further enhance LLM capabilities.

Looking forward, ongoing research and innovation are essential to unlock the full potential of LLMs within IoT ecosystems. This survey serves as a comprehensive and foundational resource for researchers and practitioners, offering insights to drive advancements at the intersection of these technologies. By addressing the identified gaps and focusing on cross-layer interactions, the community can enable a new generation of intelligent, efficient, and secure IoT applications.

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